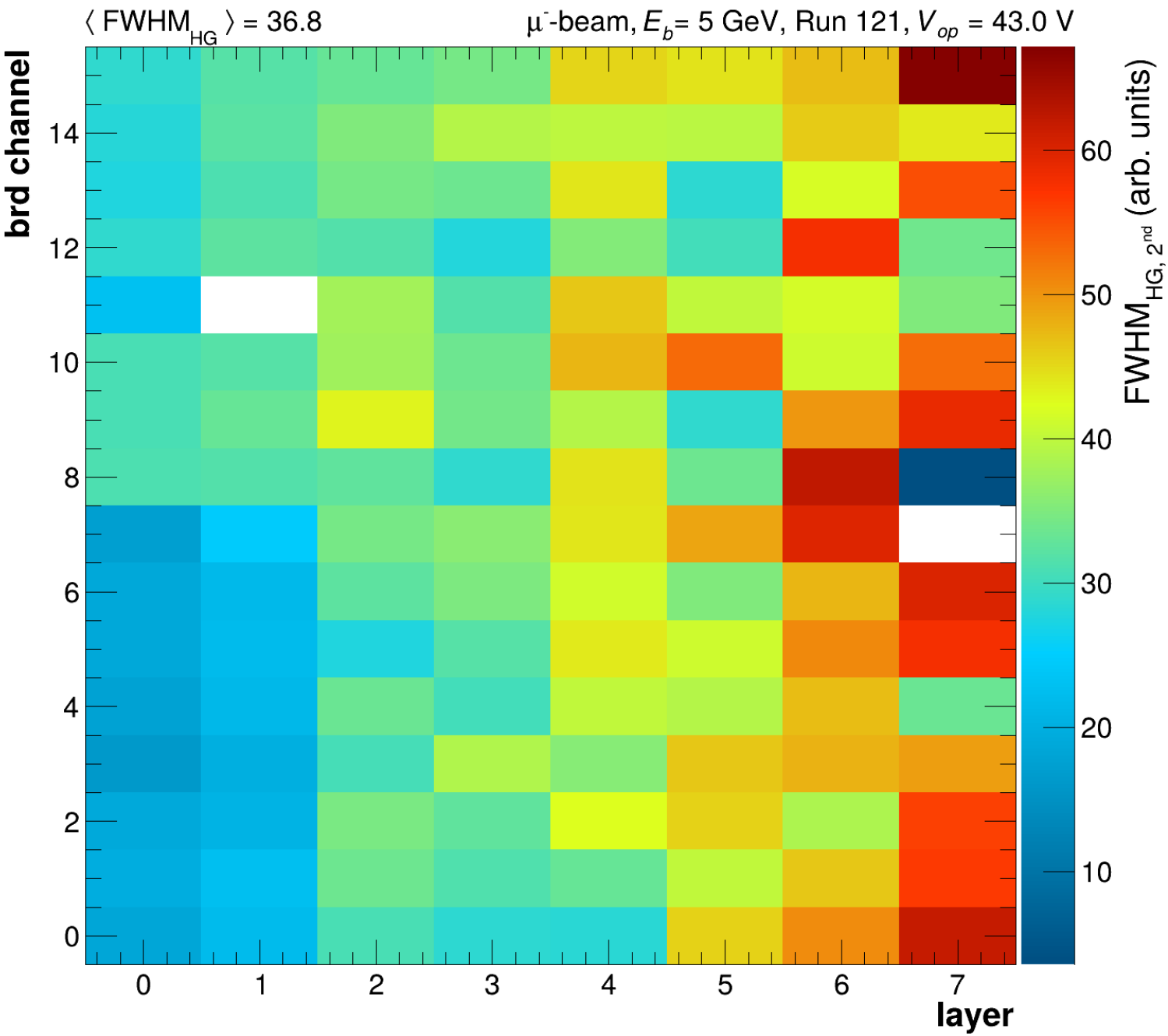
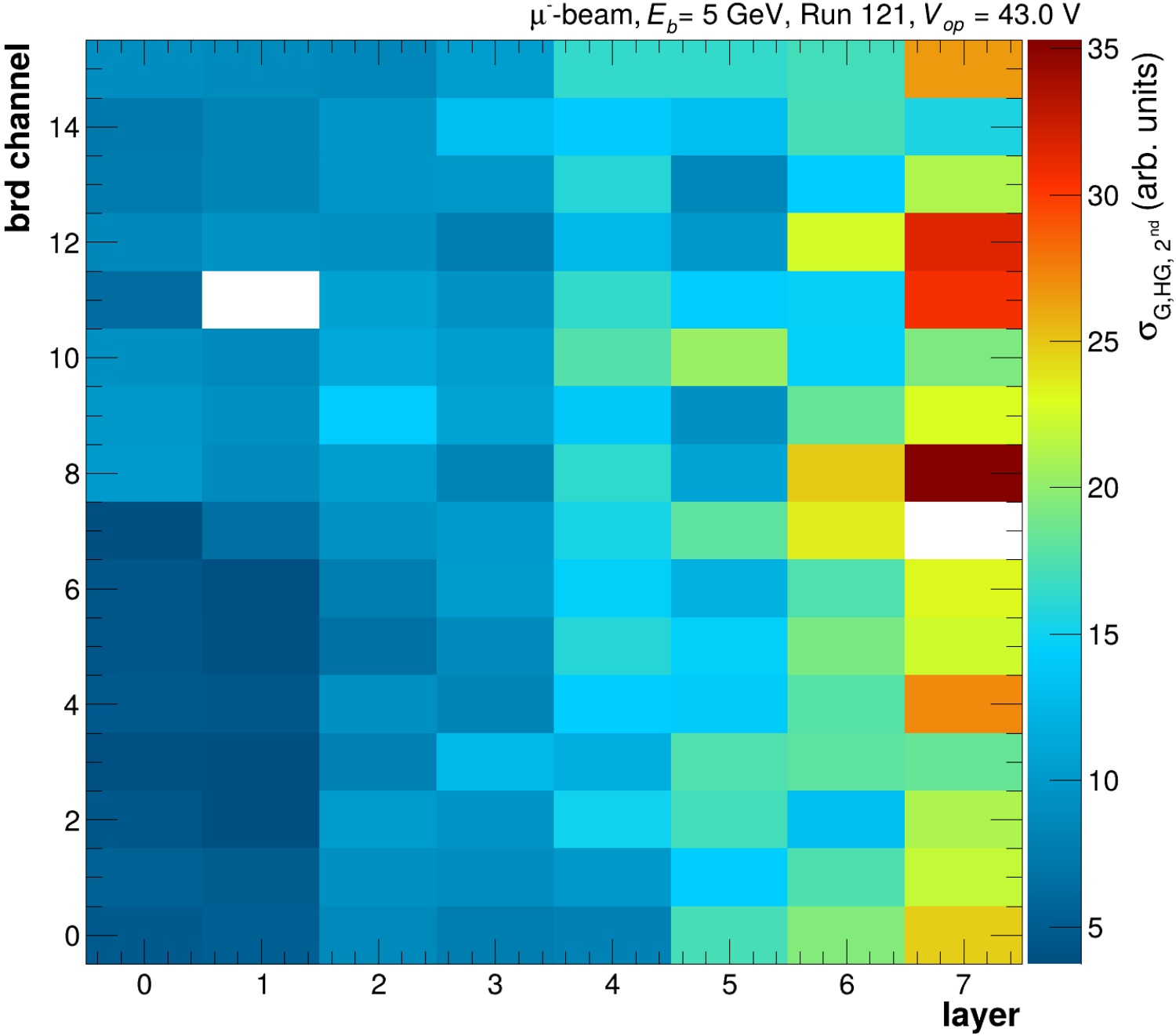
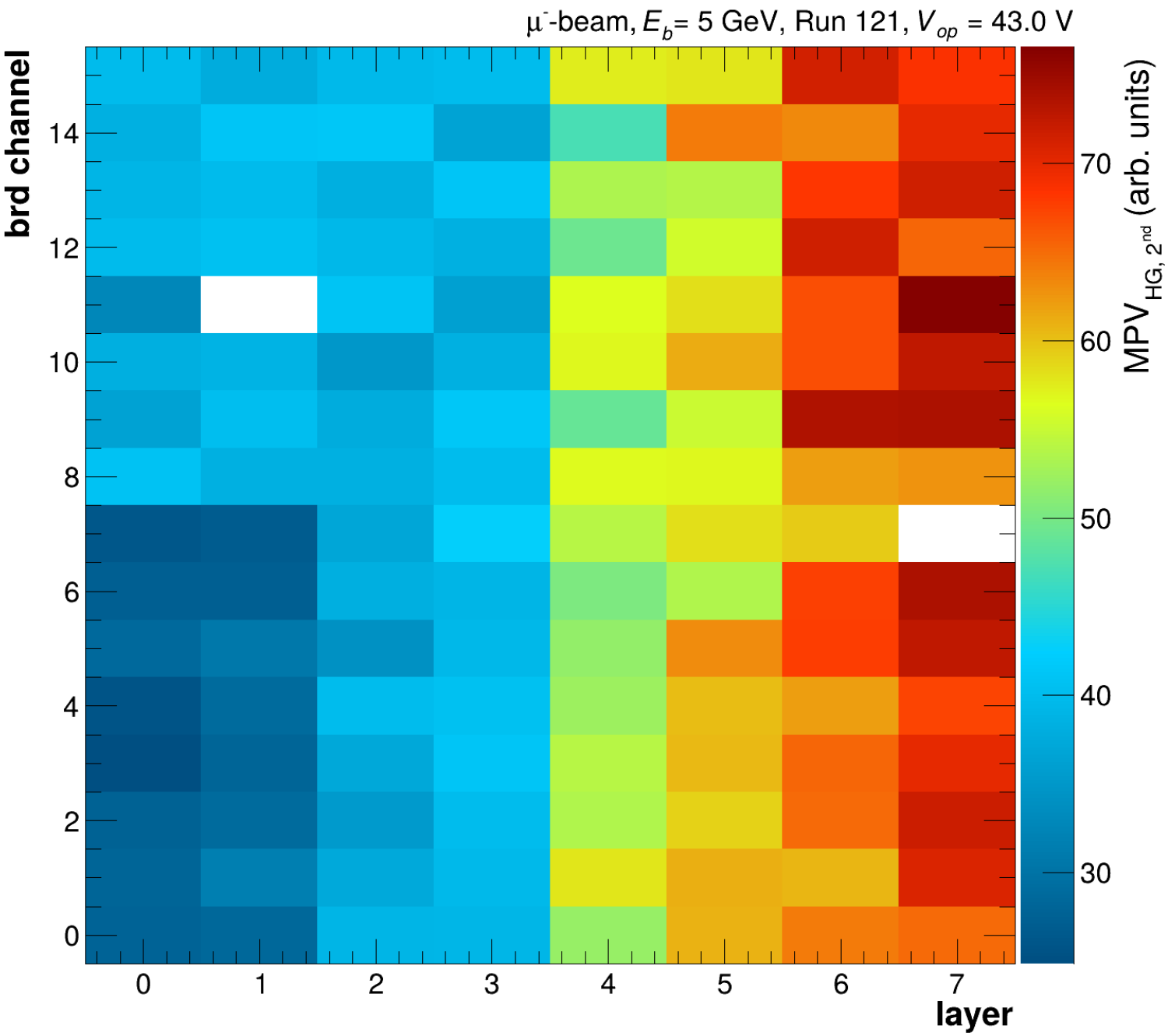
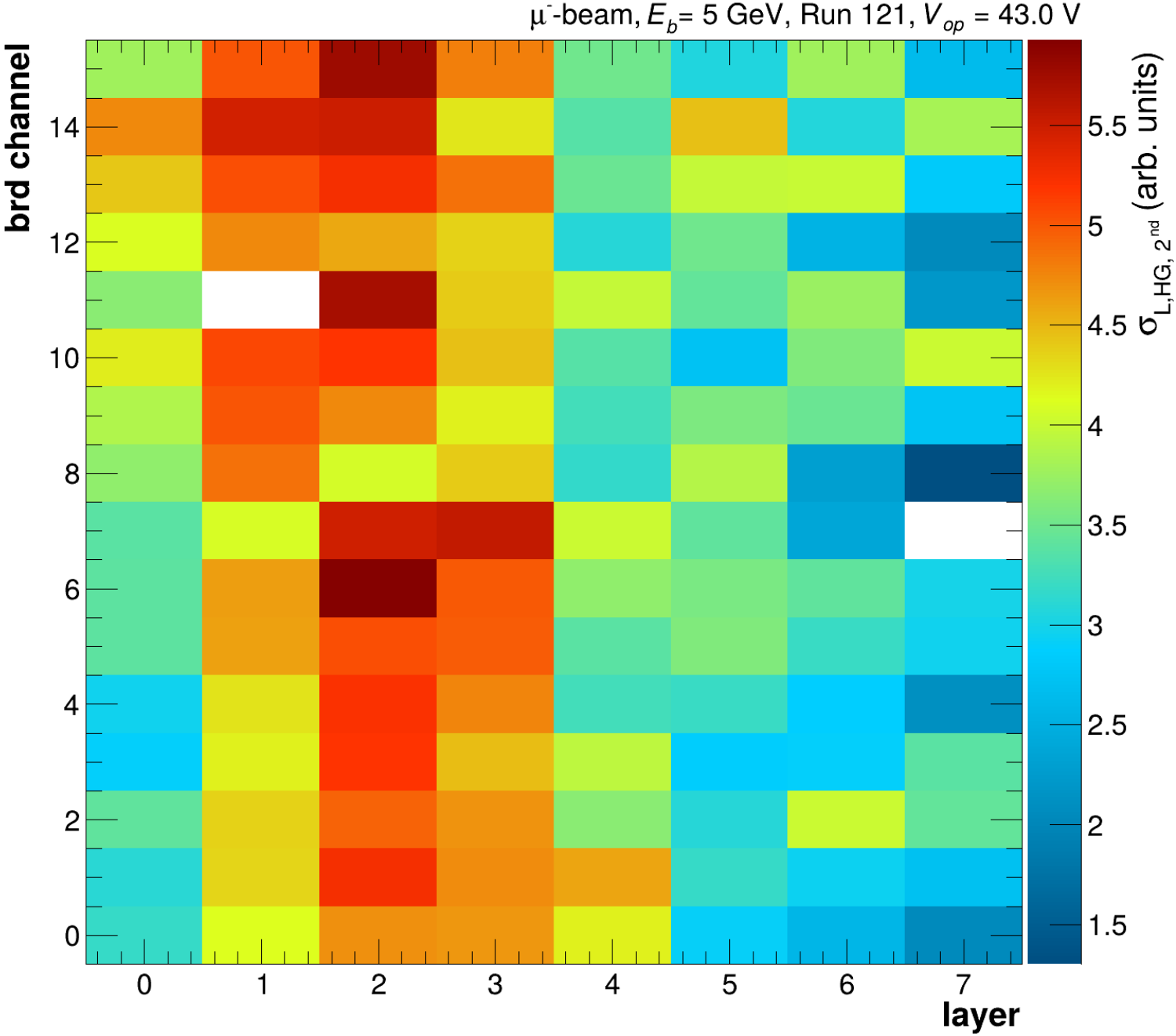
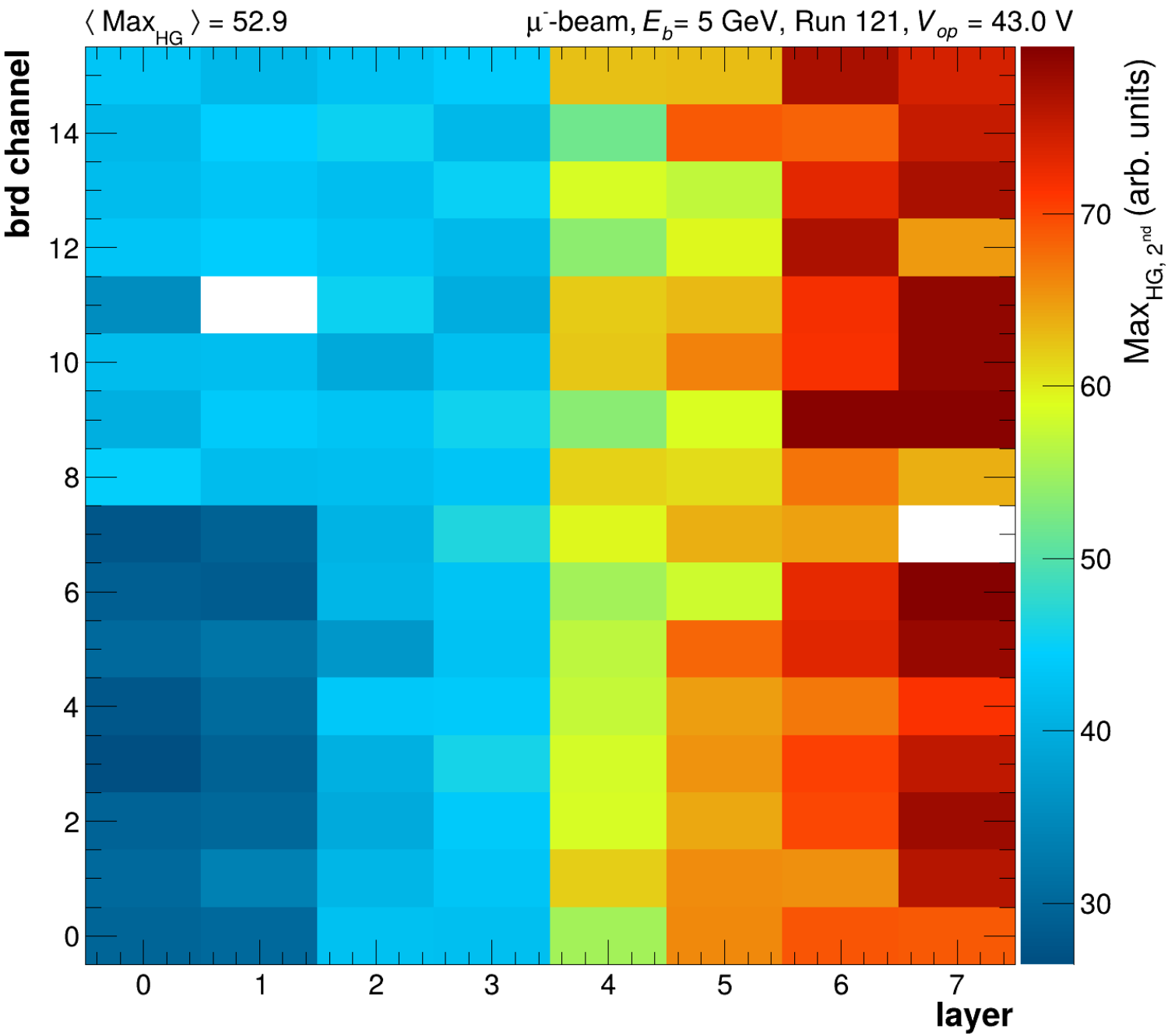
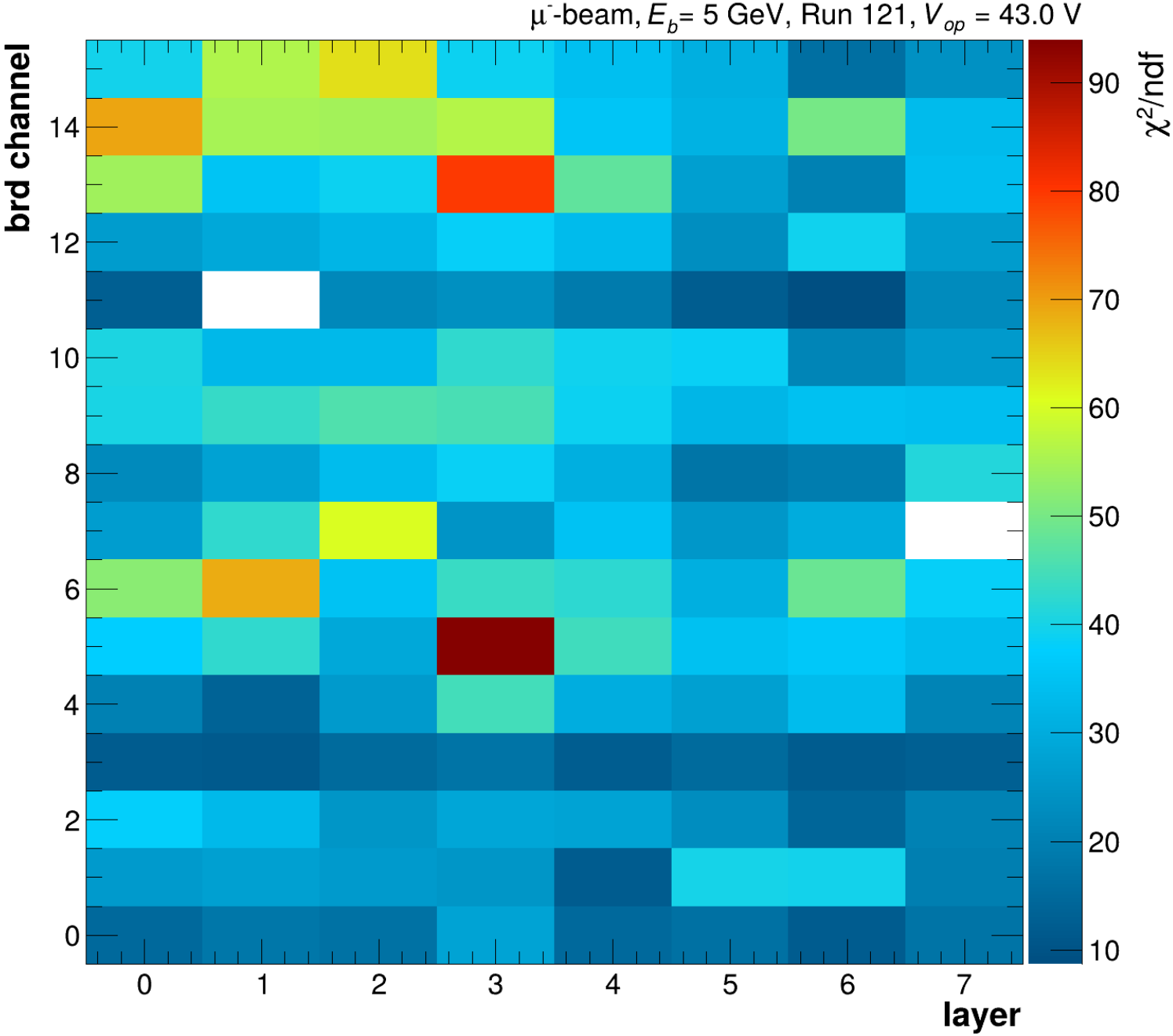

⟨ FWHM_HG ⟩ = 36.8
μ⁻-beam, E_b = 5 GeV, Run 121, V_op = 43.0 V
brd channel
layer
FWHM_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
20
30
40
50
60

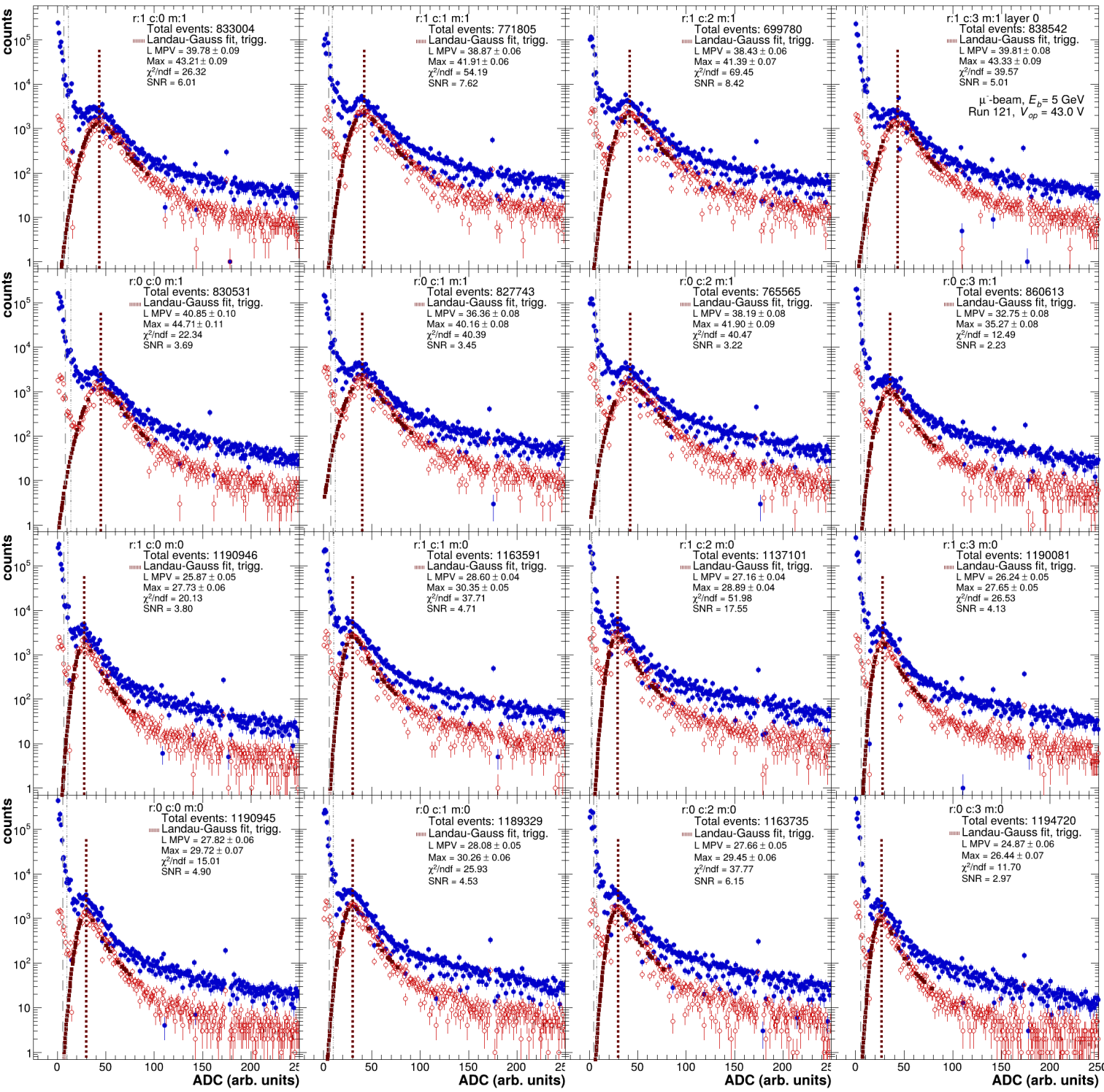

μ⁻-beam, E_b= 5 GeV, Run 121, V_{op} = 43.0 V
brd channel
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
5
10
15
20
25
30
35

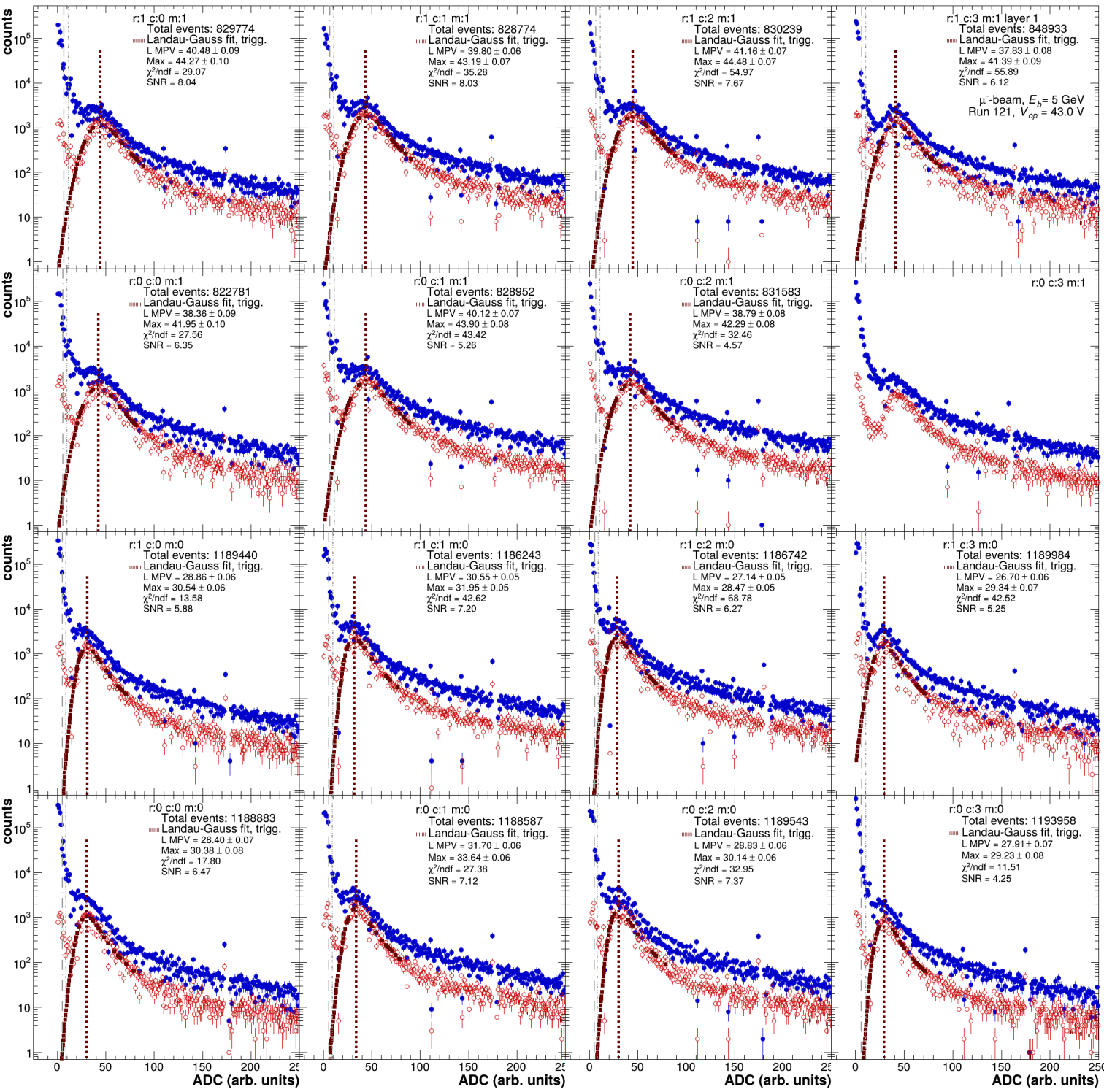

μ⁻-beam, E_b= 5 GeV, Run 121, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
30
40
50
60
70
$MPV_{HG,\ 2^{nd}}$ (arb. units)

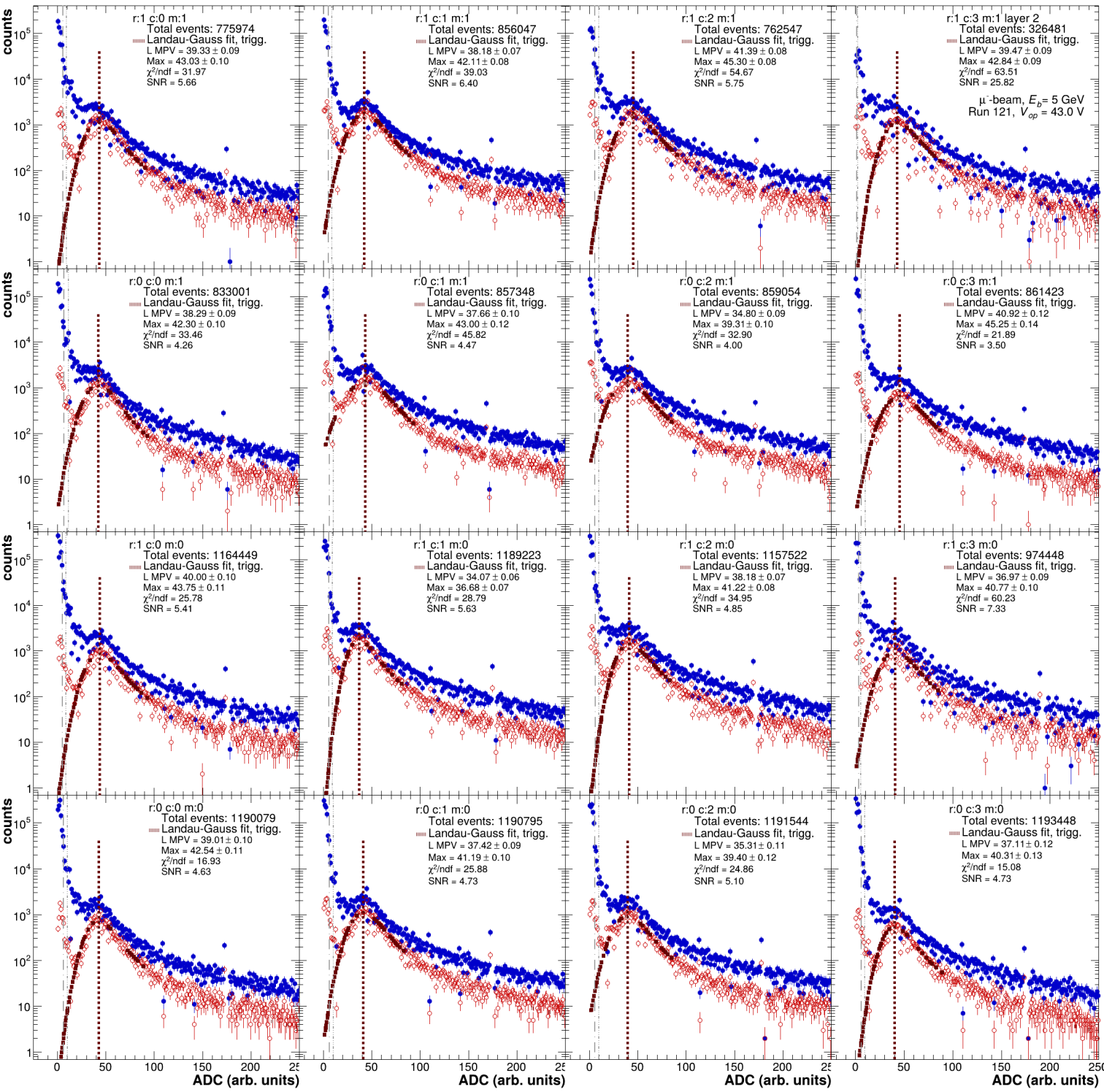

μ⁻-beam, E_b= 5 GeV, Run 121, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
1.5
2
2.5
3
3.5
4
4.5
5
5.5
$\sigma_{L,HG,2^{nd}}$ (arb. units)

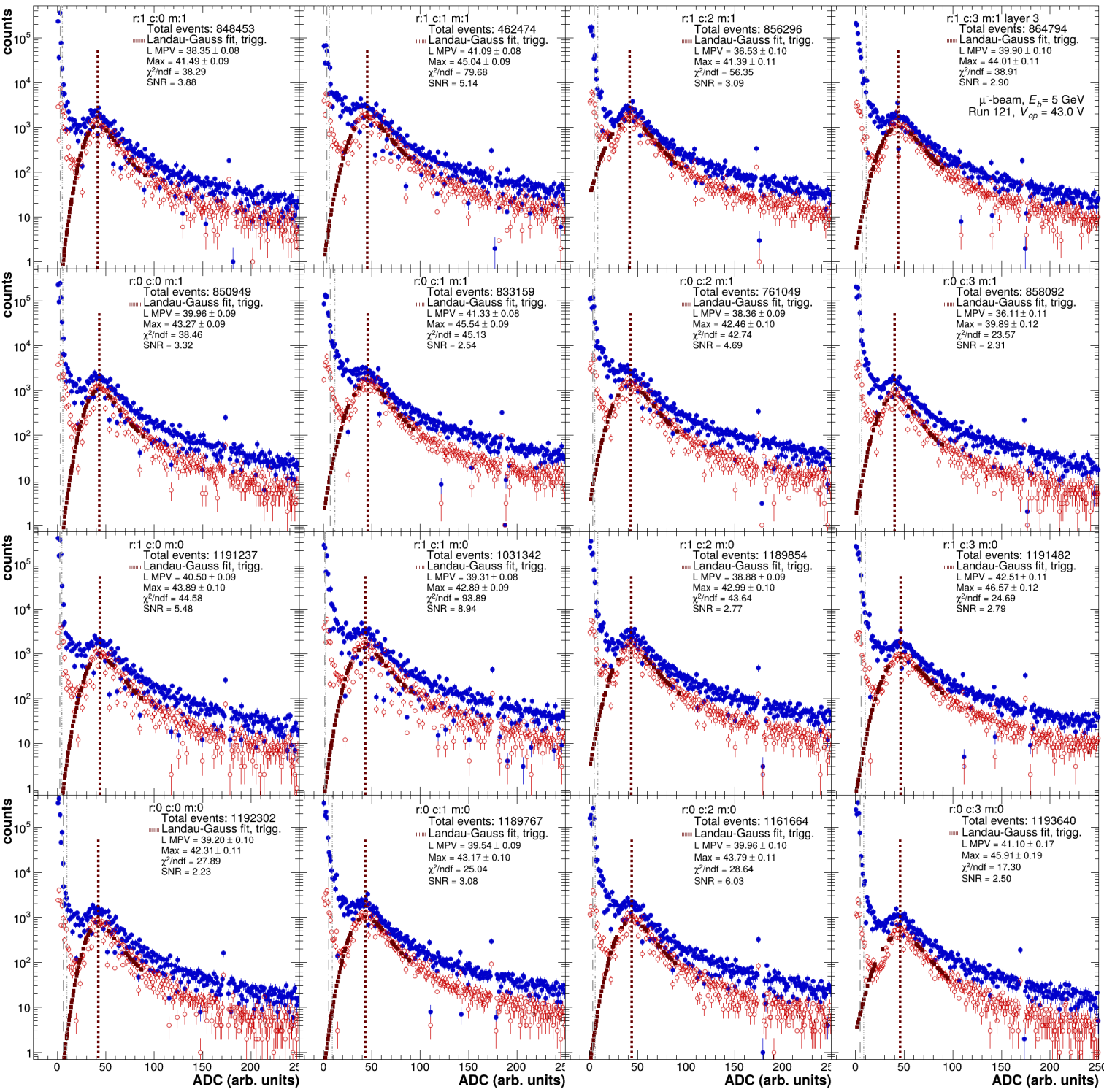

⟨ Max_HG ⟩ = 52.9
μ⁻-beam, E_b = 5 GeV, Run 121, V_op = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
Max_HG, 2nd (arb. units)
30
40
50
60
70

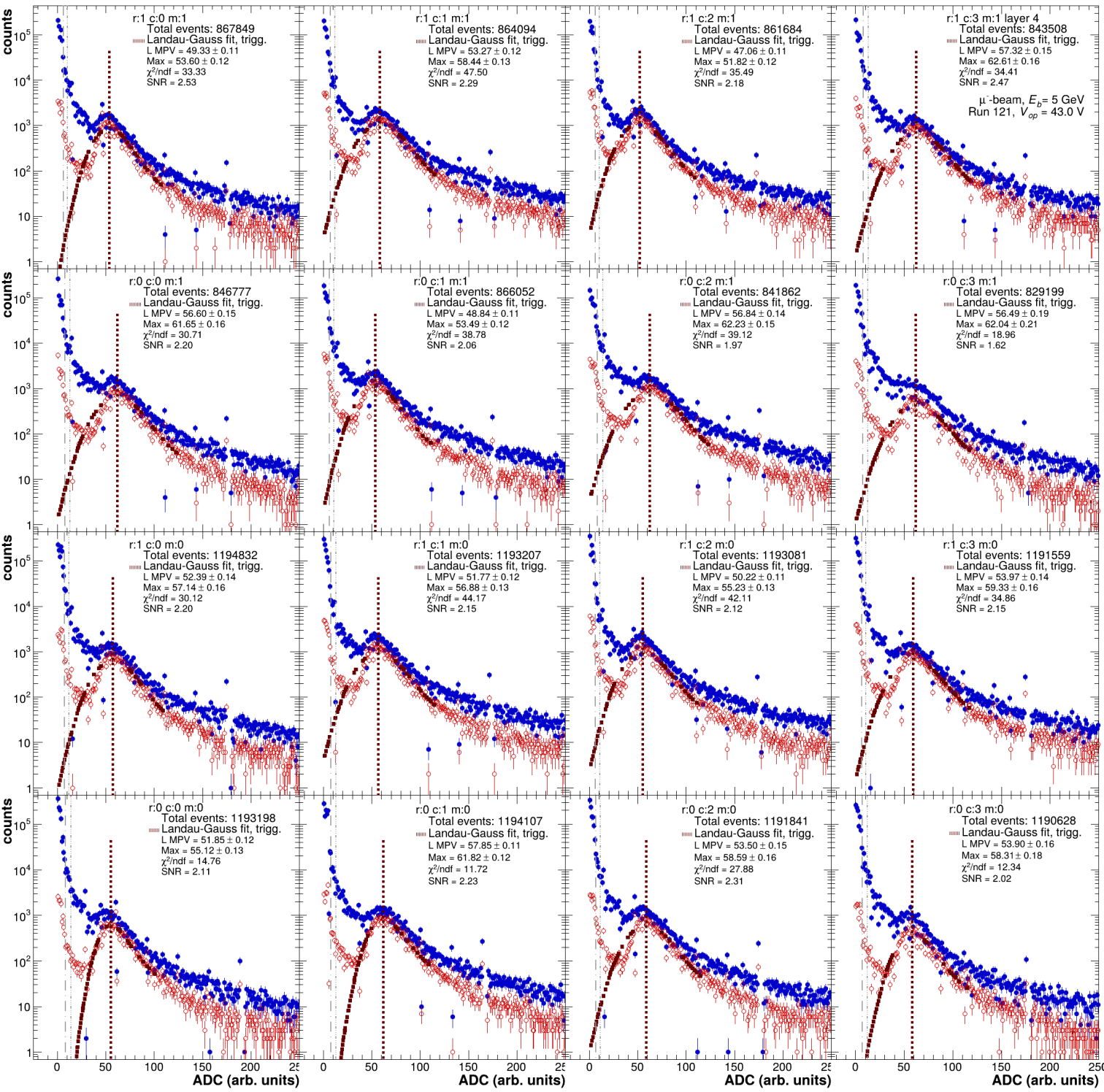

μ⁻-beam, E_b= 5 GeV, Run 121, V_{op} = 43.0 V
brd channel
layer
χ^2/ndf
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
20
30
40
50
60
70
80
90

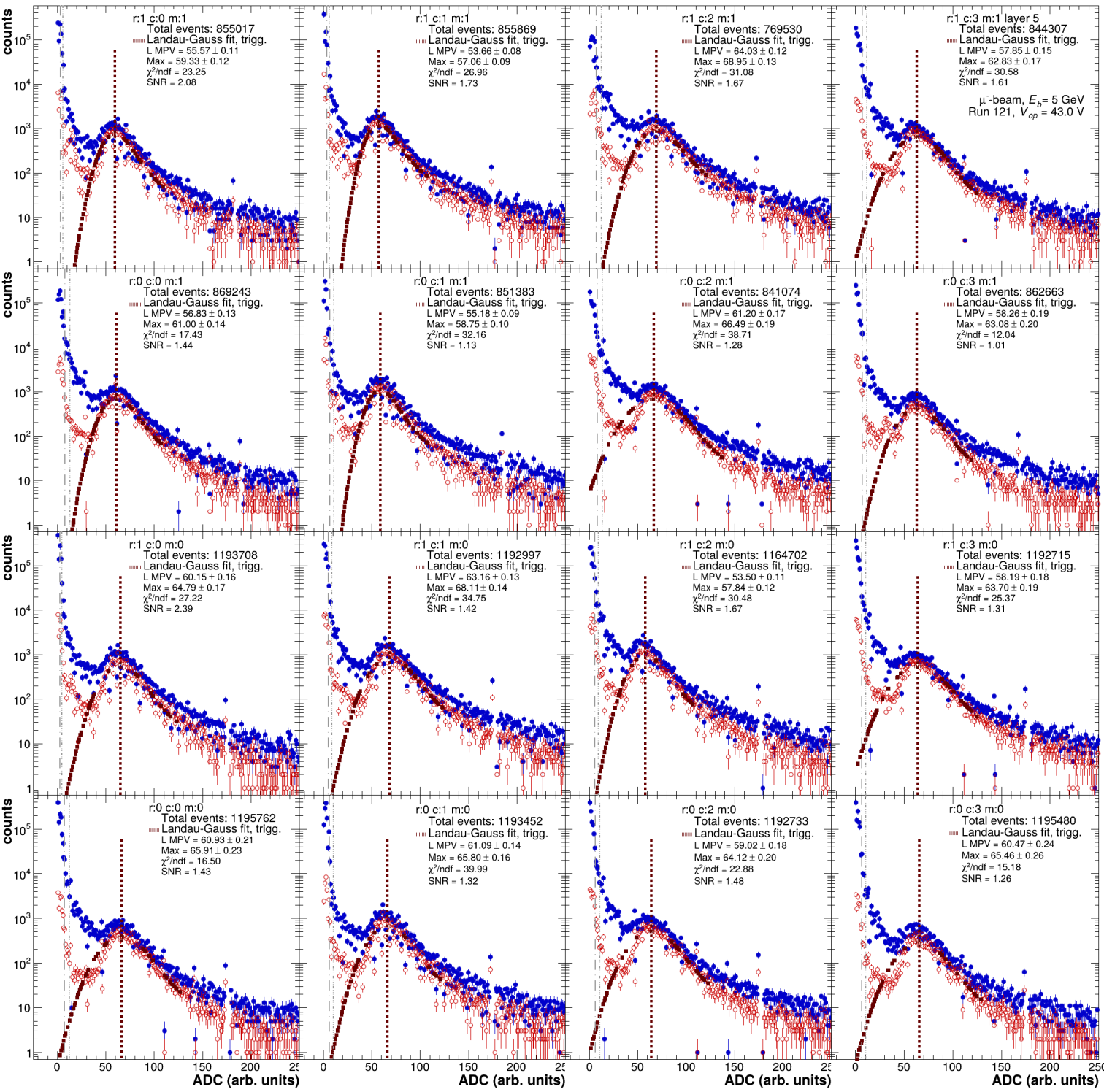

r:1 c:0 m:1
Total events: 833004
Landau-Gauss fit, trigg.
L MPV = 39.78 ± 0.09
Max = 43.21 ± 0.09
χ²/ndf = 26.32
SNR = 6.01
r:1 c:1 m:1
Total events: 771805
Landau-Gauss fit, trigg.
L MPV = 38.87 ± 0.06
Max = 41.91 ± 0.06
χ²/ndf = 54.19
SNR = 7.62
r:1 c:2 m:1
Total events: 699780
Landau-Gauss fit, trigg.
L MPV = 38.43 ± 0.06
Max = 41.39 ± 0.07
χ²/ndf = 69.45
SNR = 8.42
r:1 c:3 m:1 layer 0
Total events: 838542
Landau-Gauss fit, trigg.
L MPV = 39.81 ± 0.08
Max = 43.33 ± 0.09
χ²/ndf = 39.57
SNR = 5.01
μ⁻-beam, E_b = 5 GeV
Run 121, V_op = 43.0 V
r:0 c:0 m:1
Total events: 830531
Landau-Gauss fit, trigg.
L MPV = 40.85 ± 0.10
Max = 44.71 ± 0.11
χ²/ndf = 22.34
SNR = 3.69
r:0 c:1 m:1
Total events: 827743
Landau-Gauss fit, trigg.
L MPV = 36.36 ± 0.08
Max = 40.16 ± 0.08
χ²/ndf = 40.39
SNR = 3.45
r:0 c:2 m:1
Total events: 765565
Landau-Gauss fit, trigg.
L MPV = 38.19 ± 0.08
Max = 41.90 ± 0.09
χ²/ndf = 40.47
SNR = 3.22
r:0 c:3 m:1
Total events: 860613
Landau-Gauss fit, trigg.
L MPV = 32.75 ± 0.08
Max = 35.27 ± 0.08
χ²/ndf = 12.49
SNR = 2.23
r:1 c:0 m:0
Total events: 1190946
Landau-Gauss fit, trigg.
L MPV = 25.87 ± 0.05
Max = 27.73 ± 0.06
χ²/ndf = 20.13
SNR = 3.80
r:1 c:1 m:0
Total events: 1163591
Landau-Gauss fit, trigg.
L MPV = 28.60 ± 0.04
Max = 30.35 ± 0.05
χ²/ndf = 37.71
SNR = 4.71
r:1 c:2 m:0
Total events: 1137101
Landau-Gauss fit, trigg.
L MPV = 27.16 ± 0.04
Max = 28.89 ± 0.04
χ²/ndf = 51.98
SNR = 17.55
r:1 c:3 m:0
Total events: 1190081
Landau-Gauss fit, trigg.
L MPV = 26.24 ± 0.05
Max = 27.65 ± 0.05
χ²/ndf = 26.53
SNR = 4.13
r:0 c:0 m:0
Total events: 1190945
Landau-Gauss fit, trigg.
L MPV = 27.82 ± 0.06
Max = 29.72 ± 0.07
χ²/ndf = 15.01
SNR = 4.90
r:0 c:1 m:0
Total events: 1189329
Landau-Gauss fit, trigg.
L MPV = 28.08 ± 0.05
Max = 30.26 ± 0.06
χ²/ndf = 25.93
SNR = 4.53
r:0 c:2 m:0
Total events: 1163735
Landau-Gauss fit, trigg.
L MPV = 27.66 ± 0.05
Max = 29.45 ± 0.06
χ²/ndf = 37.77
SNR = 6.15
r:0 c:3 m:0
Total events: 1194720
Landau-Gauss fit, trigg.
L MPV = 24.87 ± 0.06
Max = 26.44 ± 0.07
χ²/ndf = 11.70
SNR = 2.97
counts
ADC (arb. units)

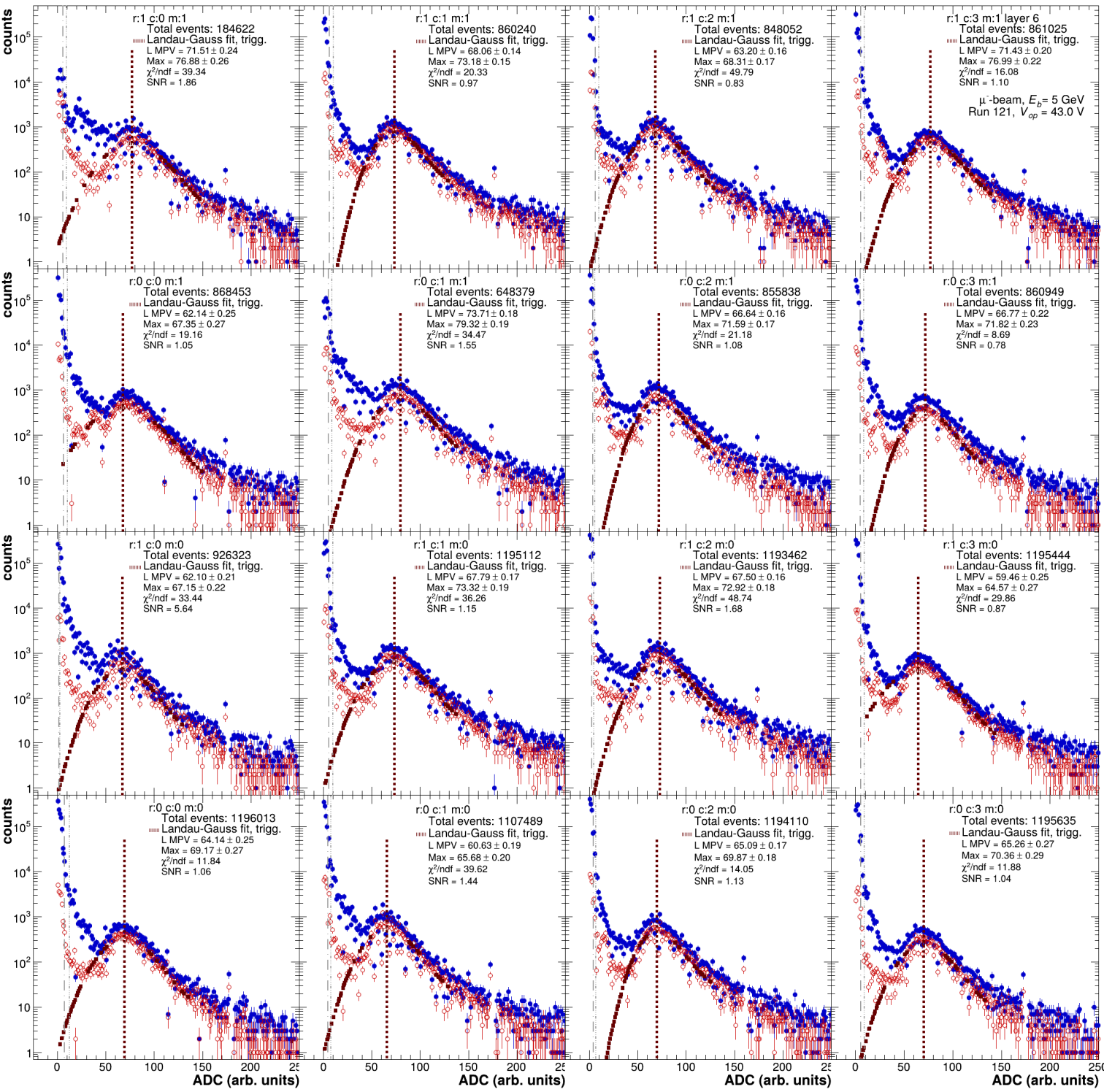

r:1 c:0 m:1
Total events: 829774
Landau-Gauss fit, trigg.
L MPV = 40.48 ± 0.09
Max = 44.27 ± 0.10
χ²/ndf = 29.07
SNR = 8.04
r:1 c:1 m:1
Total events: 828774
Landau-Gauss fit, trigg.
L MPV = 39.80 ± 0.06
Max = 43.19 ± 0.07
χ²/ndf = 35.28
SNR = 8.03
r:1 c:2 m:1
Total events: 830239
Landau-Gauss fit, trigg.
L MPV = 41.16 ± 0.07
Max = 44.48 ± 0.07
χ²/ndf = 54.97
SNR = 7.67
r:1 c:3 m:1 layer 1
Total events: 848933
Landau-Gauss fit, trigg.
L MPV = 37.83 ± 0.08
Max = 41.39 ± 0.09
χ²/ndf = 55.89
SNR = 6.12
μ-beam, E_b = 5 GeV
Run 121, V_{op} = 43.0 V
r:0 c:0 m:1
Total events: 822781
Landau-Gauss fit, trigg.
L MPV = 38.36 ± 0.09
Max = 41.95 ± 0.10
χ²/ndf = 27.56
SNR = 6.35
r:0 c:1 m:1
Total events: 828952
Landau-Gauss fit, trigg.
L MPV = 40.12 ± 0.07
Max = 43.90 ± 0.08
χ²/ndf = 43.42
SNR = 5.26
r:0 c:2 m:1
Total events: 831583
Landau-Gauss fit, trigg.
L MPV = 38.79 ± 0.08
Max = 42.29 ± 0.08
χ²/ndf = 32.46
SNR = 4.57
r:0 c:3 m:1
r:1 c:0 m:0
Total events: 1189440
Landau-Gauss fit, trigg.
L MPV = 28.86 ± 0.06
Max = 30.54 ± 0.06
χ²/ndf = 13.58
SNR = 5.88
r:1 c:1 m:0
Total events: 1186243
Landau-Gauss fit, trigg.
L MPV = 30.55 ± 0.05
Max = 31.95 ± 0.05
χ²/ndf = 42.62
SNR = 7.20
r:1 c:2 m:0
Total events: 1186742
Landau-Gauss fit, trigg.
L MPV = 27.14 ± 0.05
Max = 28.47 ± 0.05
χ²/ndf = 68.78
SNR = 6.27
r:1 c:3 m:0
Total events: 1189984
Landau-Gauss fit, trigg.
L MPV = 26.70 ± 0.06
Max = 29.34 ± 0.07
χ²/ndf = 42.52
SNR = 5.25
r:0 c:0 m:0
Total events: 1188883
Landau-Gauss fit, trigg.
L MPV = 28.40 ± 0.07
Max = 30.38 ± 0.08
χ²/ndf = 17.80
SNR = 6.47
r:0 c:1 m:0
Total events: 1188587
Landau-Gauss fit, trigg.
L MPV = 31.70 ± 0.06
Max = 33.64 ± 0.06
χ²/ndf = 27.38
SNR = 7.12
r:0 c:2 m:0
Total events: 1189543
Landau-Gauss fit, trigg.
L MPV = 28.83 ± 0.06
Max = 30.14 ± 0.06
χ²/ndf = 32.95
SNR = 7.37
r:0 c:3 m:0
Total events: 1193958
Landau-Gauss fit, trigg.
L MPV = 27.91 ± 0.07
Max = 29.23 ± 0.08
χ²/ndf = 11.51
SNR = 4.25
counts
ADC (arb. units)

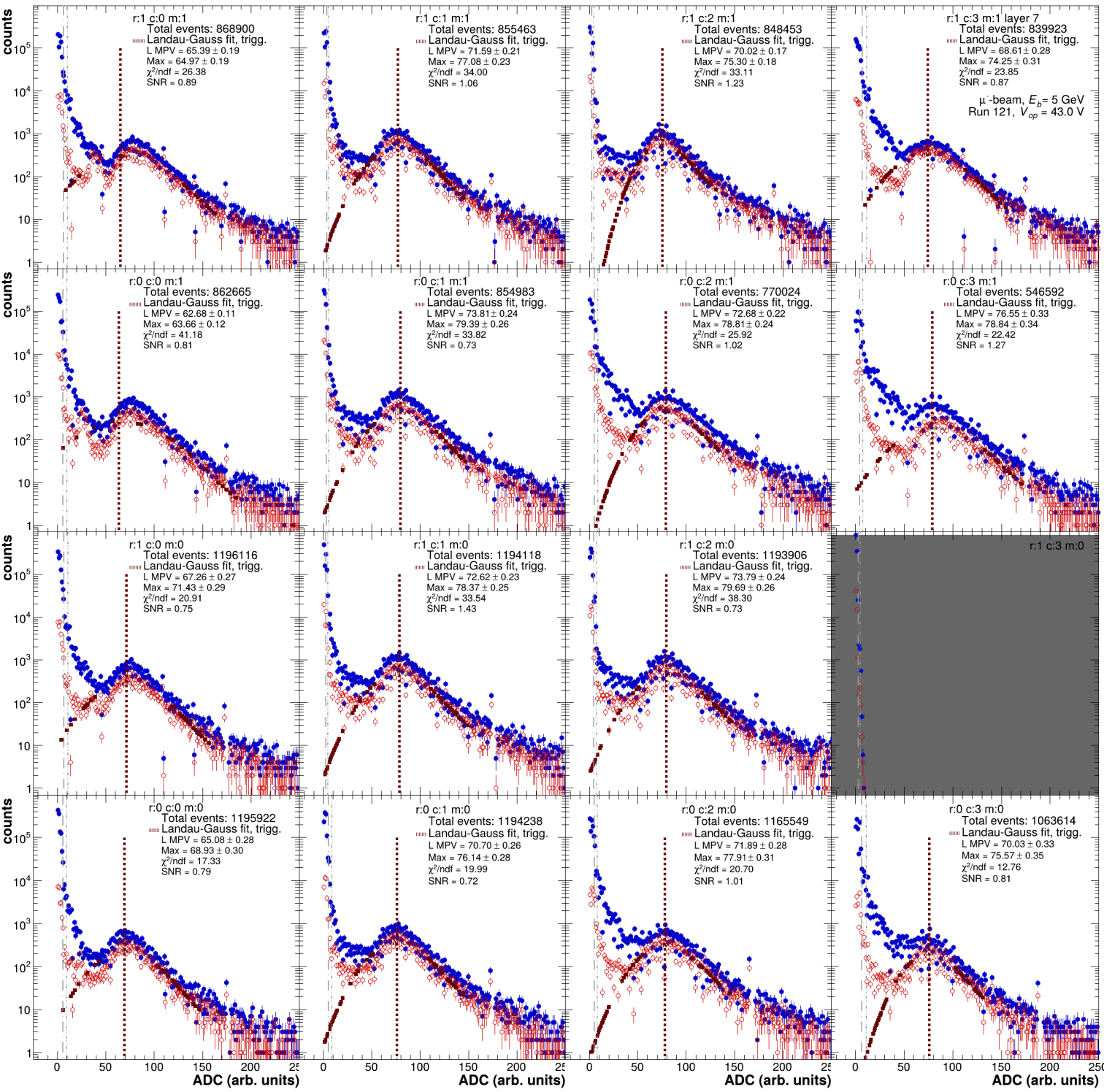

r:1 c:0 m:1
Total events: 775974
Landau-Gauss fit, trigg.
L MPV = 39.33 ± 0.09
Max = 43.03 ± 0.10
χ²/ndf = 31.97
SNR = 5.66
r:1 c:1 m:1
Total events: 856047
Landau-Gauss fit, trigg.
L MPV = 38.18 ± 0.07
Max = 42.11 ± 0.08
χ²/ndf = 39.03
SNR = 6.40
r:1 c:2 m:1
Total events: 762547
Landau-Gauss fit, trigg.
L MPV = 41.39 ± 0.08
Max = 45.30 ± 0.08
χ²/ndf = 54.67
SNR = 5.75
r:1 c:3 m:1 layer 2
Total events: 326481
Landau-Gauss fit, trigg.
L MPV = 39.47 ± 0.09
Max = 42.84 ± 0.09
χ²/ndf = 63.51
SNR = 25.82
μ⁻-beam, E_b = 5 GeV
Run 121, V_op = 43.0 V
r:0 c:0 m:1
Total events: 833001
Landau-Gauss fit, trigg.
L MPV = 38.29 ± 0.09
Max = 42.30 ± 0.10
χ²/ndf = 33.46
SNR = 4.26
r:0 c:1 m:1
Total events: 857348
Landau-Gauss fit, trigg.
L MPV = 37.66 ± 0.10
Max = 43.00 ± 0.12
χ²/ndf = 45.82
SNR = 4.47
r:0 c:2 m:1
Total events: 859054
Landau-Gauss fit, trigg.
L MPV = 34.80 ± 0.09
Max = 39.31 ± 0.10
χ²/ndf = 32.90
SNR = 4.00
r:0 c:3 m:1
Total events: 861423
Landau-Gauss fit, trigg.
L MPV = 40.92 ± 0.12
Max = 45.25 ± 0.14
χ²/ndf = 21.89
SNR = 3.50
r:1 c:0 m:0
Total events: 1164449
Landau-Gauss fit, trigg.
L MPV = 40.00 ± 0.10
Max = 43.75 ± 0.11
χ²/ndf = 25.78
SNR = 5.41
r:1 c:1 m:0
Total events: 1189223
Landau-Gauss fit, trigg.
L MPV = 34.07 ± 0.06
Max = 36.68 ± 0.07
χ²/ndf = 28.79
SNR = 5.63
r:1 c:2 m:0
Total events: 1157522
Landau-Gauss fit, trigg.
L MPV = 38.18 ± 0.07
Max = 41.22 ± 0.08
χ²/ndf = 34.95
SNR = 4.85
r:1 c:3 m:0
Total events: 974448
Landau-Gauss fit, trigg.
L MPV = 36.97 ± 0.09
Max = 40.77 ± 0.10
χ²/ndf = 60.23
SNR = 7.33
r:0 c:0 m:0
Total events: 1190079
Landau-Gauss fit, trigg.
L MPV = 39.01 ± 0.10
Max = 42.54 ± 0.11
χ²/ndf = 16.93
SNR = 4.63
r:0 c:1 m:0
Total events: 1190795
Landau-Gauss fit, trigg.
L MPV = 37.42 ± 0.09
Max = 41.19 ± 0.10
χ²/ndf = 25.88
SNR = 4.73
r:0 c:2 m:0
Total events: 1191544
Landau-Gauss fit, trigg.
L MPV = 35.31 ± 0.11
Max = 39.40 ± 0.12
χ²/ndf = 24.86
SNR = 5.10
r:0 c:3 m:0
Total events: 1193448
Landau-Gauss fit, trigg.
L MPV = 37.11 ± 0.12
Max = 40.31 ± 0.13
χ²/ndf = 15.08
SNR = 4.73
counts
ADC (arb. units)

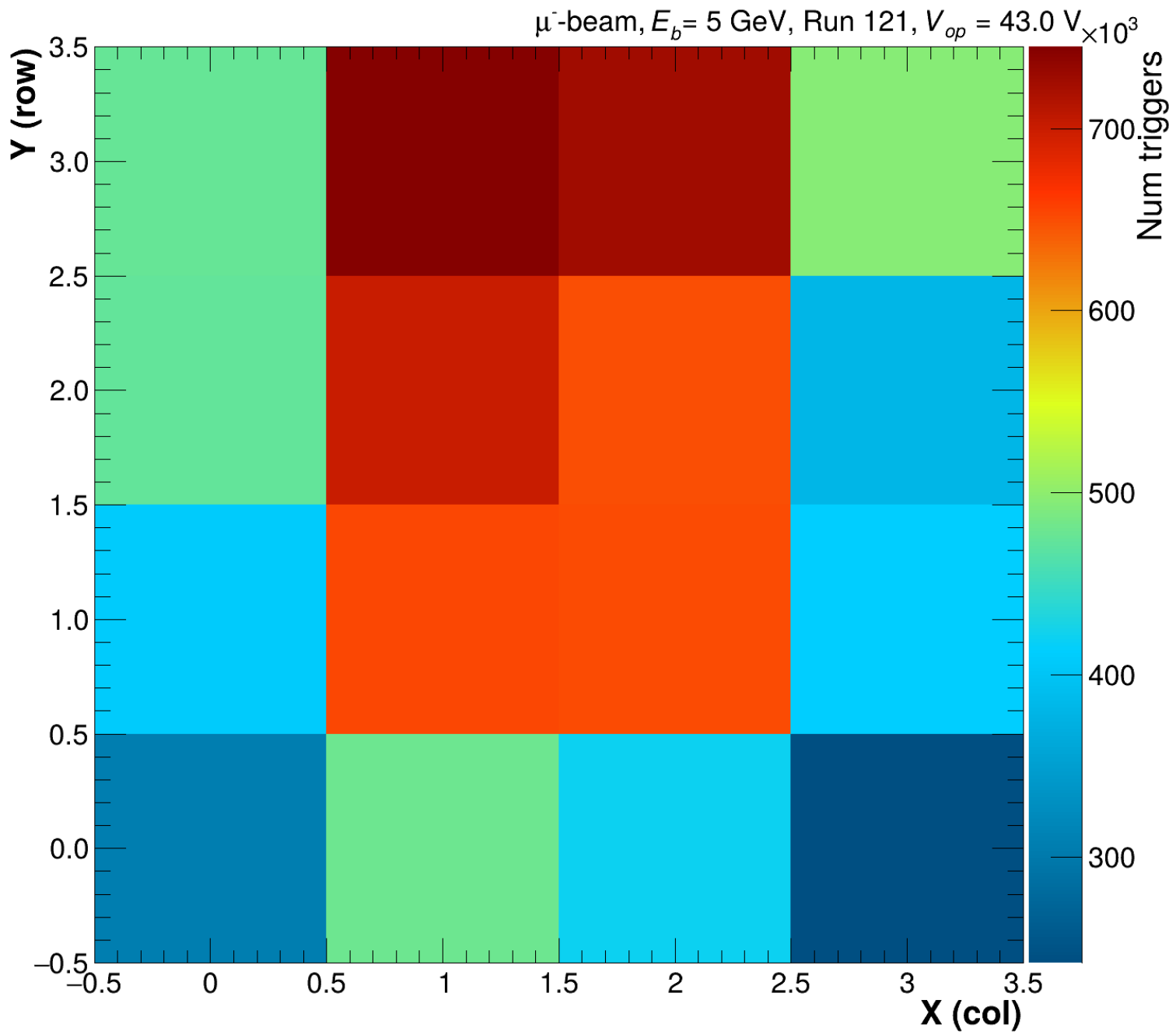

r:1 c:0 m:1
Total events: 848453
Landau-Gauss fit, trigg.
L MPV = 38.35 ± 0.08
Max = 41.49 ± 0.09
χ²/ndf = 38.29
SNR = 3.88
r:1 c:1 m:1
Total events: 462474
Landau-Gauss fit, trigg.
L MPV = 41.09 ± 0.08
Max = 45.04 ± 0.09
χ²/ndf = 79.68
SNR = 5.14
r:1 c:2 m:1
Total events: 856296
Landau-Gauss fit, trigg.
L MPV = 36.53 ± 0.10
Max = 41.39 ± 0.11
χ²/ndf = 56.35
SNR = 3.09
r:1 c:3 m:1 layer 3
Total events: 864794
Landau-Gauss fit, trigg.
L MPV = 39.90 ± 0.10
Max = 44.01 ± 0.11
χ²/ndf = 38.91
SNR = 2.90
μ-beam, E_b = 5 GeV
Run 121, V_op = 43.0 V
r:0 c:0 m:1
Total events: 850949
Landau-Gauss fit, trigg.
L MPV = 39.96 ± 0.09
Max = 43.27 ± 0.09
χ²/ndf = 38.46
SNR = 3.32
r:0 c:1 m:1
Total events: 833159
Landau-Gauss fit, trigg.
L MPV = 41.33 ± 0.08
Max = 45.54 ± 0.09
χ²/ndf = 45.13
SNR = 2.54
r:0 c:2 m:1
Total events: 761049
Landau-Gauss fit, trigg.
L MPV = 38.36 ± 0.09
Max = 42.46 ± 0.10
χ²/ndf = 42.74
SNR = 4.69
r:0 c:3 m:1
Total events: 858092
Landau-Gauss fit, trigg.
L MPV = 36.11 ± 0.11
Max = 39.89 ± 0.12
χ²/ndf = 23.57
SNR = 2.31
r:1 c:0 m:0
Total events: 1191237
Landau-Gauss fit, trigg.
L MPV = 40.50 ± 0.09
Max = 43.89 ± 0.10
χ²/ndf = 44.58
SNR = 5.48
r:1 c:1 m:0
Total events: 1031342
Landau-Gauss fit, trigg.
L MPV = 39.31 ± 0.08
Max = 42.89 ± 0.09
χ²/ndf = 93.89
SNR = 8.94
r:1 c:2 m:0
Total events: 1189854
Landau-Gauss fit, trigg.
L MPV = 38.88 ± 0.09
Max = 42.99 ± 0.10
χ²/ndf = 43.64
SNR = 2.77
r:1 c:3 m:0
Total events: 1191482
Landau-Gauss fit, trigg.
L MPV = 42.51 ± 0.11
Max = 46.57 ± 0.12
χ²/ndf = 24.69
SNR = 2.79
r:0 c:0 m:0
Total events: 1192302
Landau-Gauss fit, trigg.
L MPV = 39.20 ± 0.10
Max = 42.31 ± 0.11
χ²/ndf = 27.89
SNR = 2.23
r:0 c:1 m:0
Total events: 1189767
Landau-Gauss fit, trigg.
L MPV = 39.54 ± 0.09
Max = 43.17 ± 0.10
χ²/ndf = 25.04
SNR = 3.08
r:0 c:2 m:0
Total events: 1161664
Landau-Gauss fit, trigg.
L MPV = 39.96 ± 0.10
Max = 43.79 ± 0.11
χ²/ndf = 28.64
SNR = 6.03
r:0 c:3 m:0
Total events: 1193640
Landau-Gauss fit, trigg.
L MPV = 41.10 ± 0.17
Max = 45.91 ± 0.19
χ²/ndf = 17.30
SNR = 2.50
counts
ADC (arb. units)

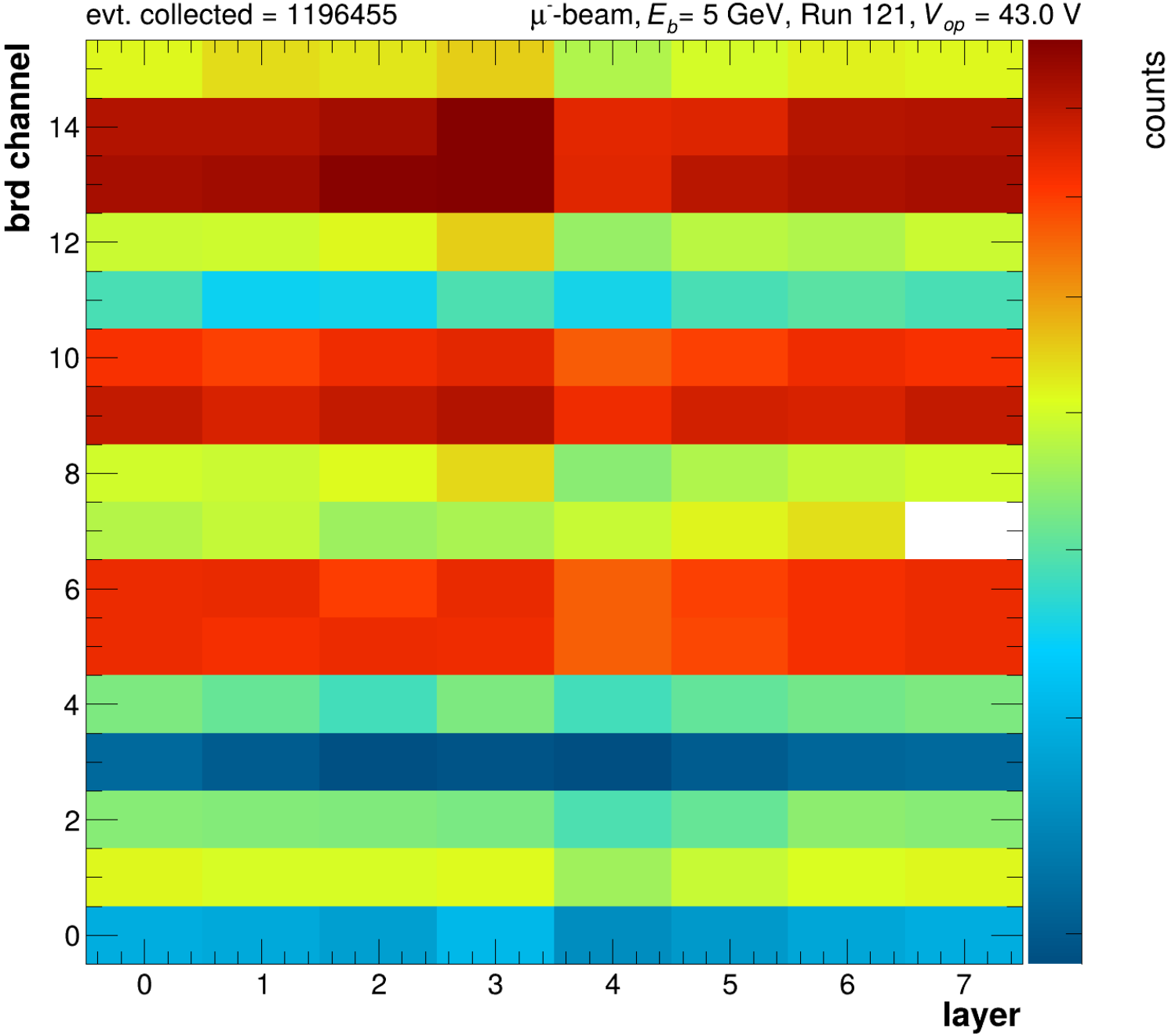

r:1 c:0 m:1
Total events: 867849
Landau-Gauss fit, trigg.
L MPV = 49.33 ± 0.11
Max = 53.60 ± 0.12
χ²/ndf = 33.33
SNR = 2.53
r:1 c:1 m:1
Total events: 864094
Landau-Gauss fit, trigg.
L MPV = 53.27 ± 0.12
Max = 58.44 ± 0.13
χ²/ndf = 47.50
SNR = 2.29
r:1 c:2 m:1
Total events: 861684
Landau-Gauss fit, trigg.
L MPV = 47.06 ± 0.11
Max = 51.82 ± 0.12
χ²/ndf = 35.49
SNR = 2.18
r:1 c:3 m:1 layer 4
Total events: 843508
Landau-Gauss fit, trigg.
L MPV = 57.32 ± 0.15
Max = 62.61 ± 0.16
χ²/ndf = 34.41
SNR = 2.47
μ⁻-beam, E_b = 5 GeV
Run 121, V_op = 43.0 V
r:0 c:0 m:1
Total events: 846777
Landau-Gauss fit, trigg.
L MPV = 56.60 ± 0.15
Max = 61.65 ± 0.16
χ²/ndf = 30.71
SNR = 2.20
r:0 c:1 m:1
Total events: 866052
Landau-Gauss fit, trigg.
L MPV = 48.84 ± 0.11
Max = 53.49 ± 0.12
χ²/ndf = 38.78
SNR = 2.06
r:0 c:2 m:1
Total events: 841862
Landau-Gauss fit, trigg.
L MPV = 56.84 ± 0.14
Max = 62.23 ± 0.15
χ²/ndf = 39.12
SNR = 1.97
r:0 c:3 m:1
Total events: 829199
Landau-Gauss fit, trigg.
L MPV = 56.49 ± 0.19
Max = 62.04 ± 0.21
χ²/ndf = 18.96
SNR = 1.62
r:1 c:0 m:0
Total events: 1194832
Landau-Gauss fit, trigg.
L MPV = 52.39 ± 0.14
Max = 57.14 ± 0.16
χ²/ndf = 30.12
SNR = 2.20
r:1 c:1 m:0
Total events: 1193207
Landau-Gauss fit, trigg.
L MPV = 51.77 ± 0.12
Max = 56.88 ± 0.13
χ²/ndf = 44.17
SNR = 2.15
r:1 c:2 m:0
Total events: 1193081
Landau-Gauss fit, trigg.
L MPV = 50.22 ± 0.11
Max = 55.23 ± 0.13
χ²/ndf = 42.11
SNR = 2.12
r:1 c:3 m:0
Total events: 1191559
Landau-Gauss fit, trigg.
L MPV = 53.97 ± 0.14
Max = 59.33 ± 0.16
χ²/ndf = 34.86
SNR = 2.15
r:0 c:0 m:0
Total events: 1193198
Landau-Gauss fit, trigg.
L MPV = 51.85 ± 0.12
Max = 55.12 ± 0.13
χ²/ndf = 14.76
SNR = 2.11
r:0 c:1 m:0
Total events: 1194107
Landau-Gauss fit, trigg.
L MPV = 57.85 ± 0.11
Max = 61.82 ± 0.12
χ²/ndf = 11.72
SNR = 2.23
r:0 c:2 m:0
Total events: 1191841
Landau-Gauss fit, trigg.
L MPV = 53.50 ± 0.15
Max = 58.59 ± 0.16
χ²/ndf = 27.88
SNR = 2.31
r:0 c:3 m:0
Total events: 1190628
Landau-Gauss fit, trigg.
L MPV = 53.90 ± 0.16
Max = 58.31 ± 0.18
χ²/ndf = 12.34
SNR = 2.02
counts
ADC (arb. units)

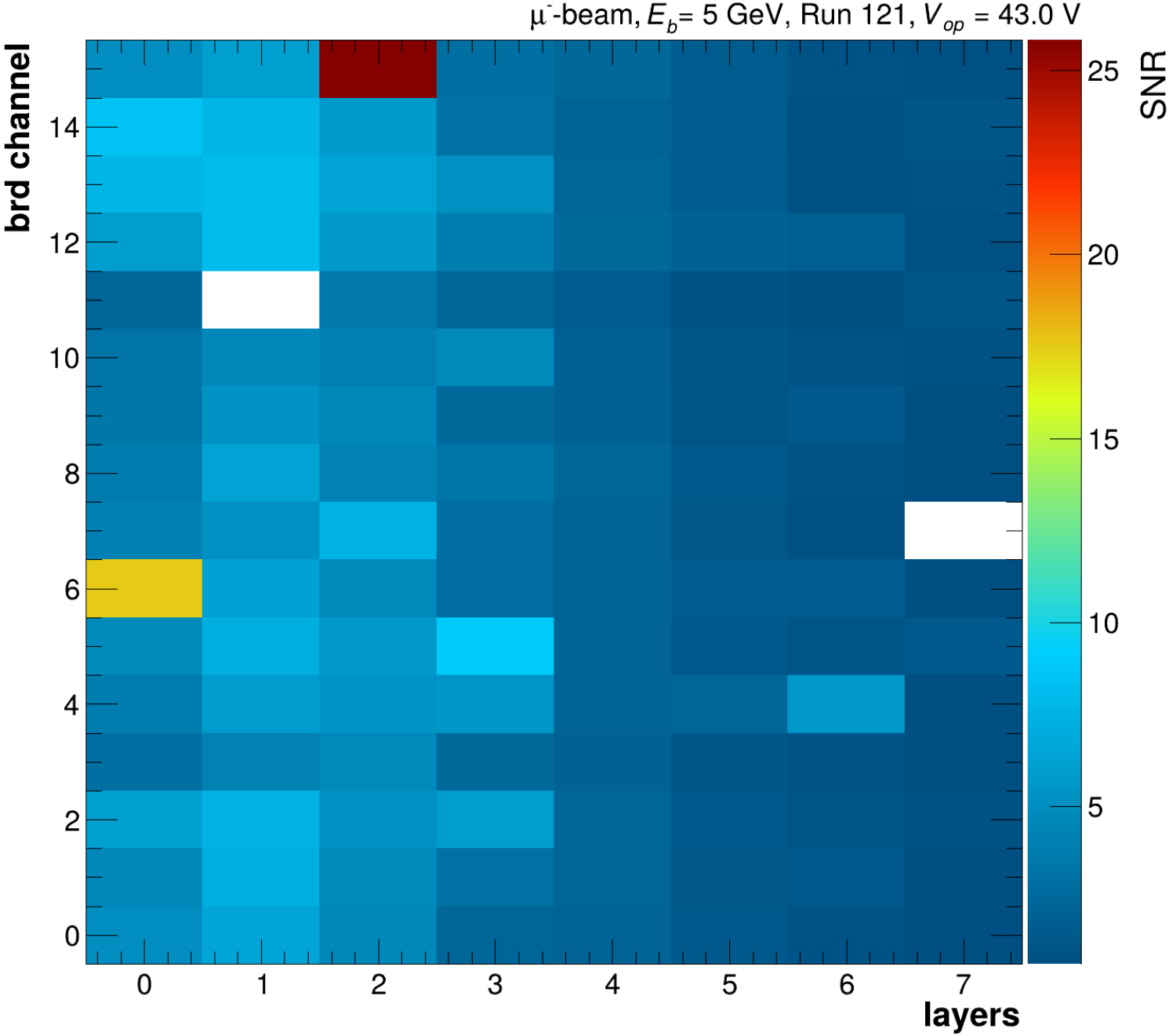

r:1 c:0 m:1
Total events: 855017
Landau-Gauss fit, trigg.
L MPV = 55.57 ± 0.11
Max = 59.33 ± 0.12
χ²/ndf = 23.25
SNR = 2.08
r:1 c:1 m:1
Total events: 855869
Landau-Gauss fit, trigg.
L MPV = 53.66 ± 0.08
Max = 57.06 ± 0.09
χ²/ndf = 26.96
SNR = 1.73
r:1 c:2 m:1
Total events: 769530
Landau-Gauss fit, trigg.
L MPV = 64.03 ± 0.12
Max = 68.95 ± 0.13
χ²/ndf = 31.08
SNR = 1.67
r:1 c:3 m:1 layer 5
Total events: 844307
Landau-Gauss fit, trigg.
L MPV = 57.85 ± 0.15
Max = 62.83 ± 0.17
χ²/ndf = 30.58
SNR = 1.61
μ⁻-beam, E_b = 5 GeV
Run 121, V_op = 43.0 V
r:0 c:0 m:1
Total events: 869243
Landau-Gauss fit, trigg.
L MPV = 56.83 ± 0.13
Max = 61.00 ± 0.14
χ²/ndf = 17.43
SNR = 1.44
r:0 c:1 m:1
Total events: 851383
Landau-Gauss fit, trigg.
L MPV = 55.18 ± 0.09
Max = 58.75 ± 0.10
χ²/ndf = 32.16
SNR = 1.13
r:0 c:2 m:1
Total events: 841074
Landau-Gauss fit, trigg.
L MPV = 61.20 ± 0.17
Max = 66.49 ± 0.19
χ²/ndf = 38.71
SNR = 1.28
r:0 c:3 m:1
Total events: 862663
Landau-Gauss fit, trigg.
L MPV = 58.26 ± 0.19
Max = 63.08 ± 0.20
χ²/ndf = 12.04
SNR = 1.01
r:1 c:0 m:0
Total events: 1193708
Landau-Gauss fit, trigg.
L MPV = 60.15 ± 0.16
Max = 64.79 ± 0.17
χ²/ndf = 27.22
SNR = 2.39
r:1 c:1 m:0
Total events: 1192997
Landau-Gauss fit, trigg.
L MPV = 63.16 ± 0.13
Max = 68.11 ± 0.14
χ²/ndf = 34.75
SNR = 1.42
r:1 c:2 m:0
Total events: 1164702
Landau-Gauss fit, trigg.
L MPV = 53.50 ± 0.11
Max = 57.84 ± 0.12
χ²/ndf = 30.48
SNR = 1.67
r:1 c:3 m:0
Total events: 1192715
Landau-Gauss fit, trigg.
L MPV = 58.19 ± 0.18
Max = 63.70 ± 0.19
χ²/ndf = 25.37
SNR = 1.31
r:0 c:0 m:0
Total events: 1195762
Landau-Gauss fit, trigg.
L MPV = 60.93 ± 0.21
Max = 65.91 ± 0.23
χ²/ndf = 16.50
SNR = 1.43
r:0 c:1 m:0
Total events: 1193452
Landau-Gauss fit, trigg.
L MPV = 61.09 ± 0.14
Max = 65.80 ± 0.16
χ²/ndf = 39.99
SNR = 1.32
r:0 c:2 m:0
Total events: 1192733
Landau-Gauss fit, trigg.
L MPV = 59.02 ± 0.18
Max = 64.12 ± 0.20
χ²/ndf = 22.88
SNR = 1.48
r:0 c:3 m:0
Total events: 1195480
Landau-Gauss fit, trigg.
L MPV = 60.47 ± 0.24
Max = 65.46 ± 0.26
χ²/ndf = 15.18
SNR = 1.26
counts
ADC (arb. units)

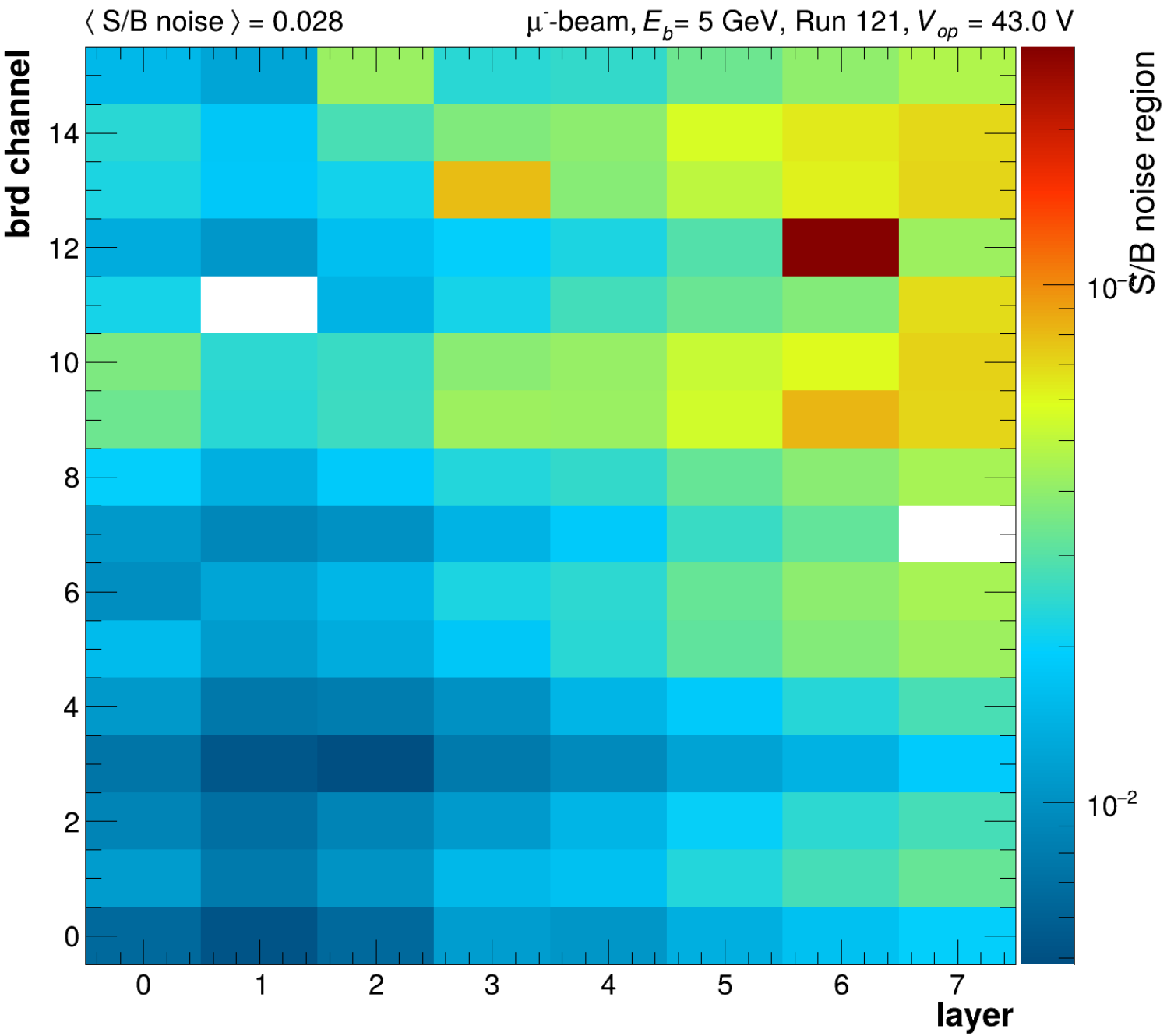

r:1 c:0 m:1
Total events: 184622
Landau-Gauss fit, trigg.
L MPV = 71.51 ± 0.24
Max = 76.88 ± 0.26
χ²/ndf = 39.34
SNR = 1.86
r:1 c:1 m:1
Total events: 860240
Landau-Gauss fit, trigg.
L MPV = 68.06 ± 0.14
Max = 73.18 ± 0.15
χ²/ndf = 20.33
SNR = 0.97
r:1 c:2 m:1
Total events: 848052
Landau-Gauss fit, trigg.
L MPV = 63.20 ± 0.16
Max = 68.31 ± 0.17
χ²/ndf = 49.79
SNR = 0.83
r:1 c:3 m:1 layer 6
Total events: 861025
Landau-Gauss fit, trigg.
L MPV = 71.43 ± 0.20
Max = 76.99 ± 0.22
χ²/ndf = 16.08
SNR = 1.10
μ⁻-beam, E_b = 5 GeV
Run 121, V_op = 43.0 V
r:0 c:0 m:1
Total events: 868453
Landau-Gauss fit, trigg.
L MPV = 62.14 ± 0.25
Max = 67.35 ± 0.27
χ²/ndf = 19.16
SNR = 1.05
r:0 c:1 m:1
Total events: 648379
Landau-Gauss fit, trigg.
L MPV = 73.71 ± 0.18
Max = 79.32 ± 0.19
χ²/ndf = 34.47
SNR = 1.55
r:0 c:2 m:1
Total events: 855838
Landau-Gauss fit, trigg.
L MPV = 66.64 ± 0.16
Max = 71.59 ± 0.17
χ²/ndf = 21.18
SNR = 1.08
r:0 c:3 m:1
Total events: 860949
Landau-Gauss fit, trigg.
L MPV = 66.77 ± 0.22
Max = 71.82 ± 0.23
χ²/ndf = 8.69
SNR = 0.78
r:1 c:0 m:0
Total events: 926323
Landau-Gauss fit, trigg.
L MPV = 62.10 ± 0.21
Max = 67.15 ± 0.22
χ²/ndf = 33.44
SNR = 5.64
r:1 c:1 m:0
Total events: 1195112
Landau-Gauss fit, trigg.
L MPV = 67.79 ± 0.17
Max = 73.32 ± 0.19
χ²/ndf = 36.26
SNR = 1.15
r:1 c:2 m:0
Total events: 1193462
Landau-Gauss fit, trigg.
L MPV = 67.50 ± 0.16
Max = 72.92 ± 0.18
χ²/ndf = 48.74
SNR = 1.68
r:1 c:3 m:0
Total events: 1195444
Landau-Gauss fit, trigg.
L MPV = 59.46 ± 0.25
Max = 64.57 ± 0.27
χ²/ndf = 29.86
SNR = 0.87
r:0 c:0 m:0
Total events: 1196013
Landau-Gauss fit, trigg.
L MPV = 64.14 ± 0.25
Max = 69.17 ± 0.27
χ²/ndf = 11.84
SNR = 1.06
r:0 c:1 m:0
Total events: 1107489
Landau-Gauss fit, trigg.
L MPV = 60.63 ± 0.19
Max = 65.68 ± 0.20
χ²/ndf = 39.62
SNR = 1.44
r:0 c:2 m:0
Total events: 1194110
Landau-Gauss fit, trigg.
L MPV = 65.09 ± 0.17
Max = 69.87 ± 0.18
χ²/ndf = 14.05
SNR = 1.13
r:0 c:3 m:0
Total events: 1195635
Landau-Gauss fit, trigg.
L MPV = 65.26 ± 0.27
Max = 70.36 ± 0.29
χ²/ndf = 11.88
SNR = 1.04
counts
ADC (arb. units)

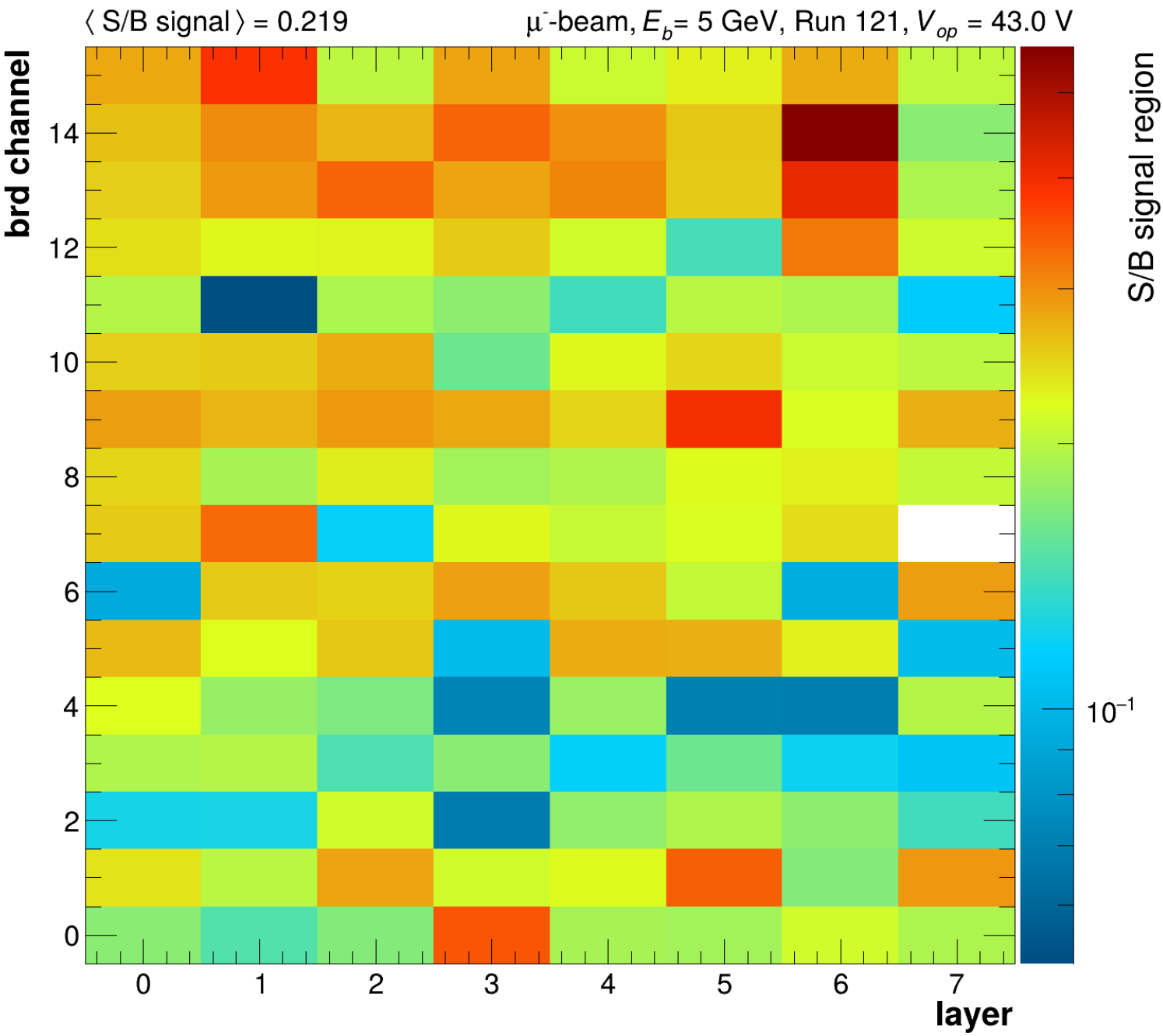

r:1 c:0 m:1
Total events: 868900
Landau-Gauss fit, trigg.
L MPV = 65.39 ± 0.19
Max = 64.97 ± 0.19
χ²/ndf = 26.38
SNR = 0.89
r:1 c:1 m:1
Total events: 855463
Landau-Gauss fit, trigg.
L MPV = 71.59 ± 0.21
Max = 77.08 ± 0.23
χ²/ndf = 34.00
SNR = 1.06
r:1 c:2 m:1
Total events: 848453
Landau-Gauss fit, trigg.
L MPV = 70.02 ± 0.17
Max = 75.30 ± 0.18
χ²/ndf = 33.11
SNR = 1.23
r:1 c:3 m:1 layer 7
Total events: 839923
Landau-Gauss fit, trigg.
L MPV = 68.61 ± 0.28
Max = 74.25 ± 0.31
χ²/ndf = 23.85
SNR = 0.87
μ⁻-beam, E_b = 5 GeV
Run 121, V_op = 43.0 V
r:0 c:0 m:1
Total events: 862665
Landau-Gauss fit, trigg.
L MPV = 62.68 ± 0.11
Max = 63.66 ± 0.12
χ²/ndf = 41.18
SNR = 0.81
r:0 c:1 m:1
Total events: 854983
Landau-Gauss fit, trigg.
L MPV = 73.81 ± 0.24
Max = 79.39 ± 0.26
χ²/ndf = 33.82
SNR = 0.73
r:0 c:2 m:1
Total events: 770024
Landau-Gauss fit, trigg.
L MPV = 72.68 ± 0.22
Max = 78.81 ± 0.24
χ²/ndf = 25.92
SNR = 1.02
r:0 c:3 m:1
Total events: 546592
Landau-Gauss fit, trigg.
L MPV = 76.55 ± 0.33
Max = 78.84 ± 0.34
χ²/ndf = 22.42
SNR = 1.27
r:1 c:0 m:0
Total events: 1196116
Landau-Gauss fit, trigg.
L MPV = 67.26 ± 0.27
Max = 71.43 ± 0.29
χ²/ndf = 20.91
SNR = 0.75
r:1 c:1 m:0
Total events: 1194118
Landau-Gauss fit, trigg.
L MPV = 72.62 ± 0.23
Max = 78.37 ± 0.25
χ²/ndf = 33.54
SNR = 1.43
r:1 c:2 m:0
Total events: 1193906
Landau-Gauss fit, trigg.
L MPV = 73.79 ± 0.24
Max = 79.69 ± 0.26
χ²/ndf = 38.30
SNR = 0.73
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 1195922
Landau-Gauss fit, trigg.
L MPV = 65.08 ± 0.28
Max = 68.93 ± 0.30
χ²/ndf = 17.33
SNR = 0.79
r:0 c:1 m:0
Total events: 1194238
Landau-Gauss fit, trigg.
L MPV = 70.70 ± 0.26
Max = 76.14 ± 0.28
χ²/ndf = 19.99
SNR = 0.72
r:0 c:2 m:0
Total events: 1165549
Landau-Gauss fit, trigg.
L MPV = 71.89 ± 0.28
Max = 77.91 ± 0.31
χ²/ndf = 20.70
SNR = 1.01
r:0 c:3 m:0
Total events: 1063614
Landau-Gauss fit, trigg.
L MPV = 70.03 ± 0.33
Max = 75.57 ± 0.35
χ²/ndf = 12.76
SNR = 0.81
counts
ADC (arb. units)

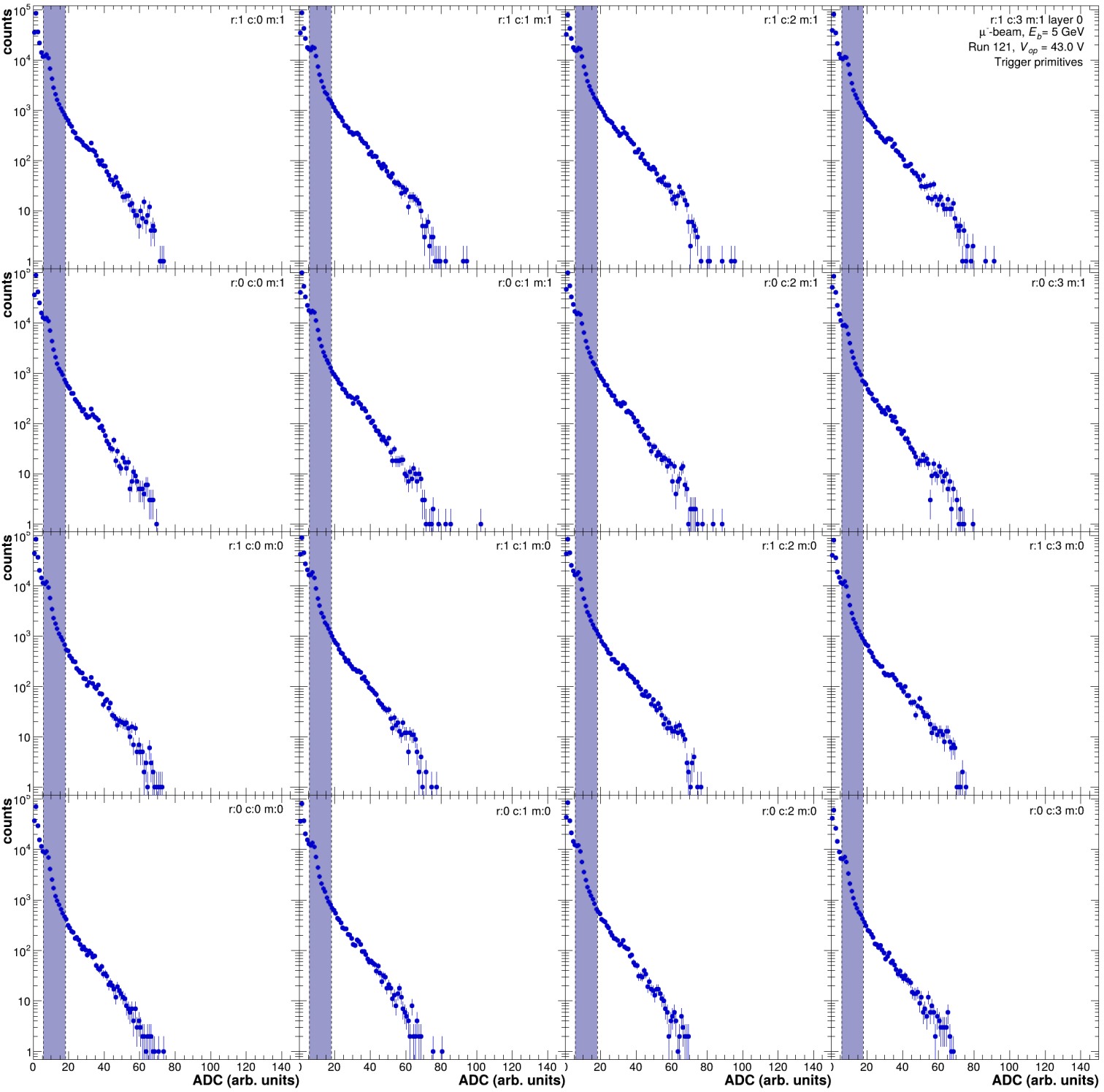

μ⁻-beam, E_b= 5 GeV, Run 121, V_{op} = 43.0 V
×10³
Num triggers
700
600
500
400
300
Y (row)
X (col)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5

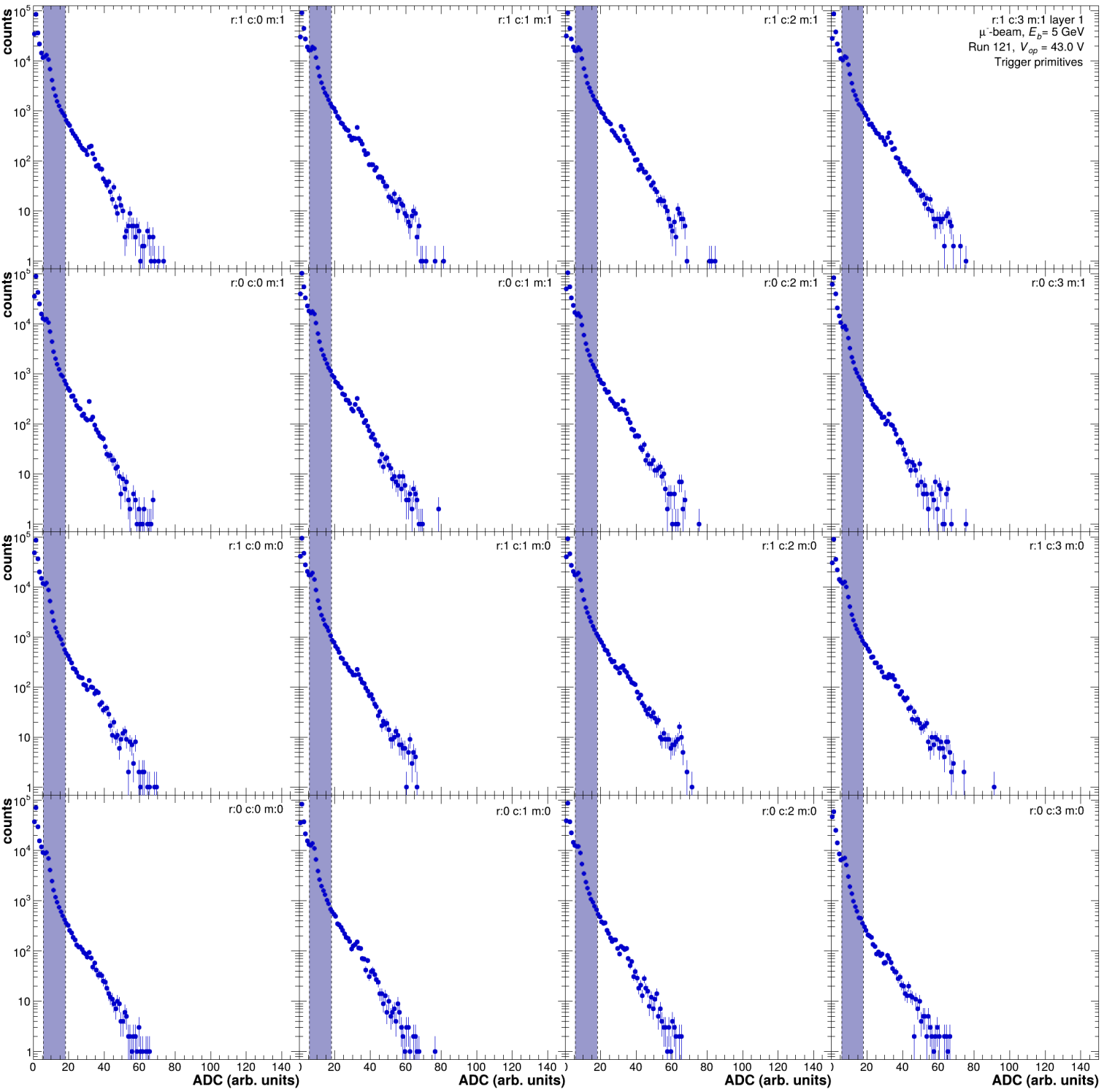

evt. collected = 1196455
μ⁻-beam, E_b= 5 GeV, Run 121, V_{op} = 43.0 V
brd channel
14
12
10
8
6
4
2
0
0
1
2
3
4
5
6
7
layer
counts

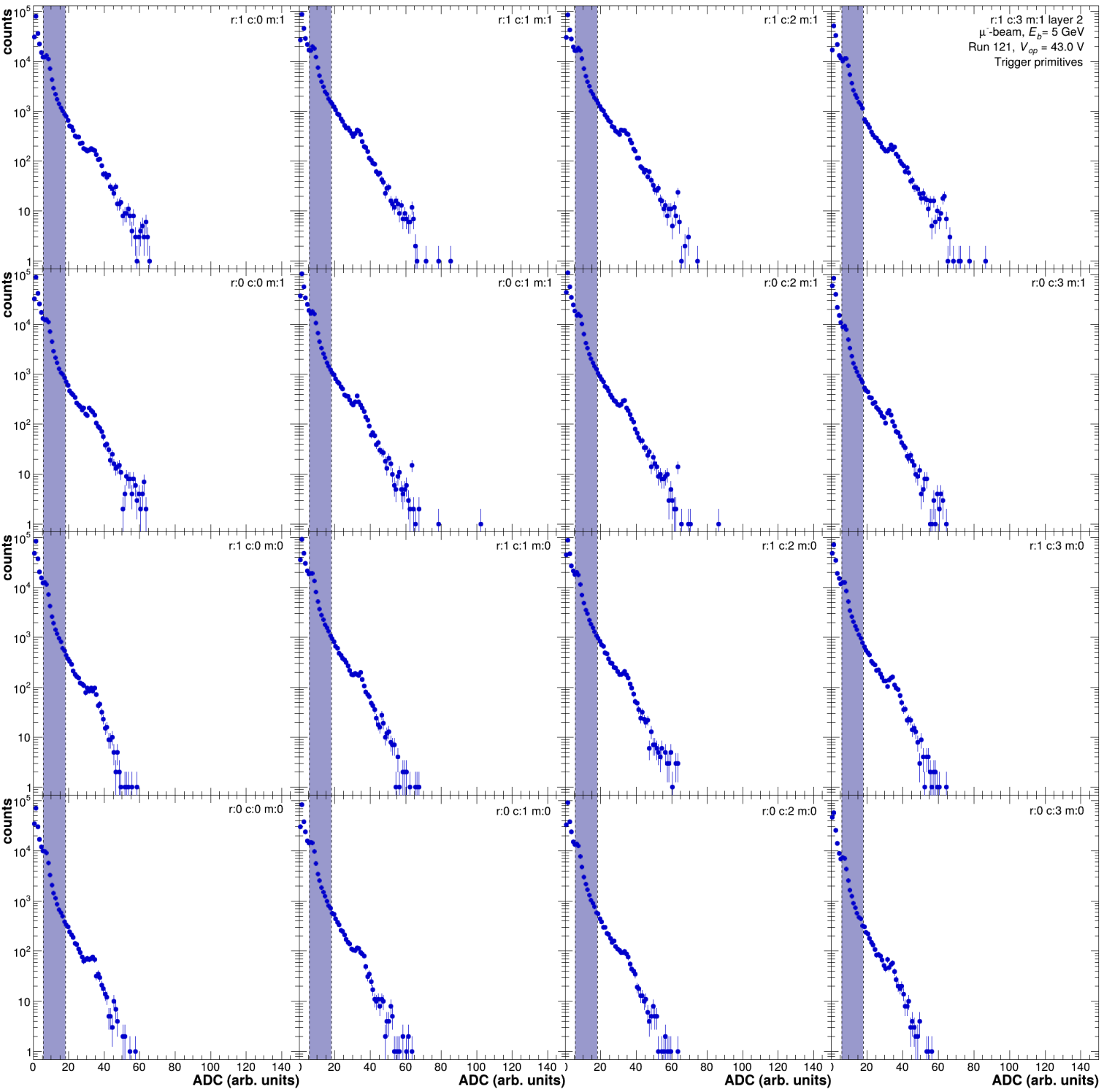

μ⁻-beam, E_b= 5 GeV, Run 121, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layers
SNR
5
10
15
20
25

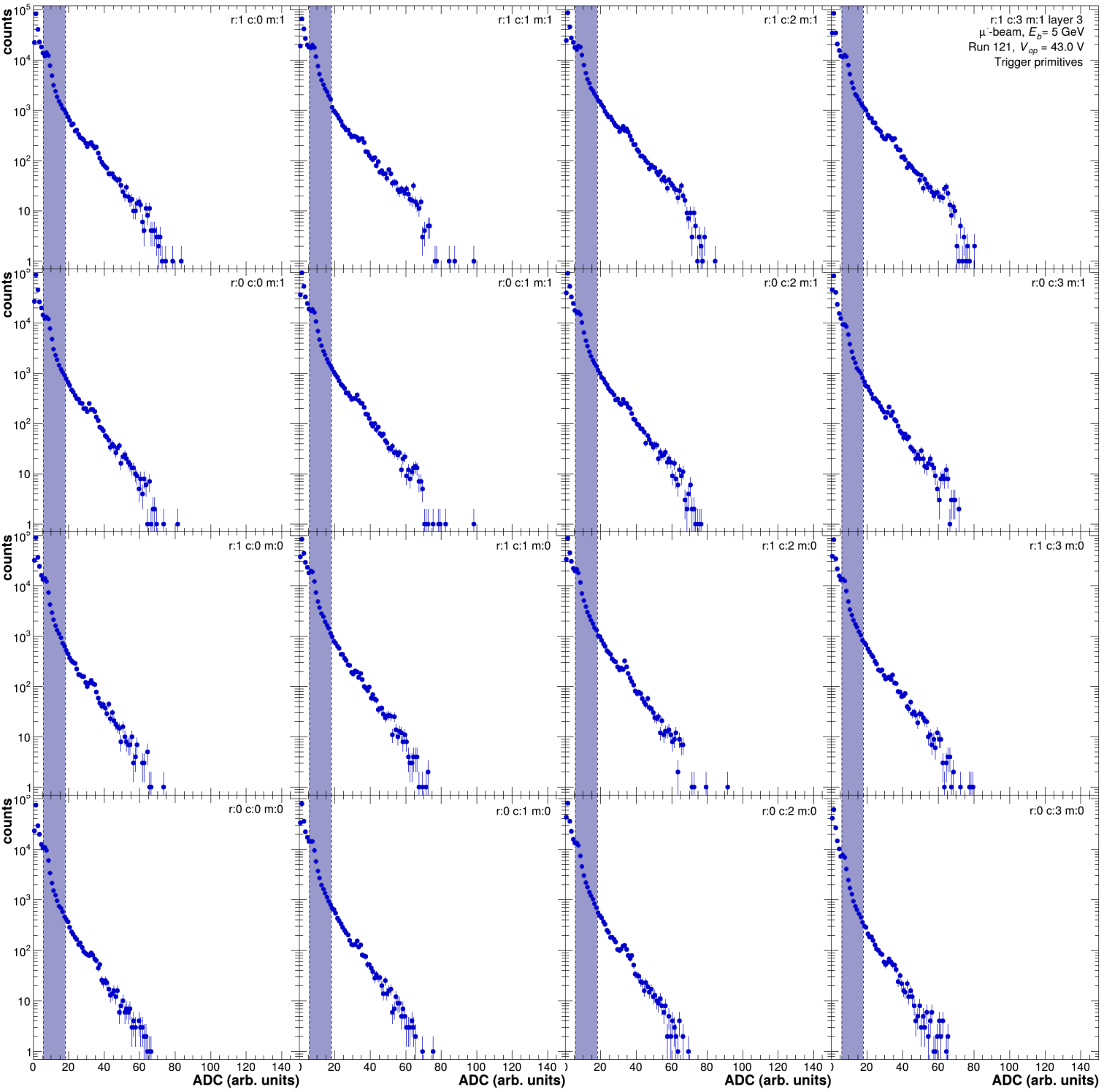

⟨ S/B noise ⟩ = 0.028
μ⁻-beam, E_b= 5 GeV, Run 121, V_{op} = 43.0 V
brd channel
layer
S/B noise region
10^{-1}
10^{-2}
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

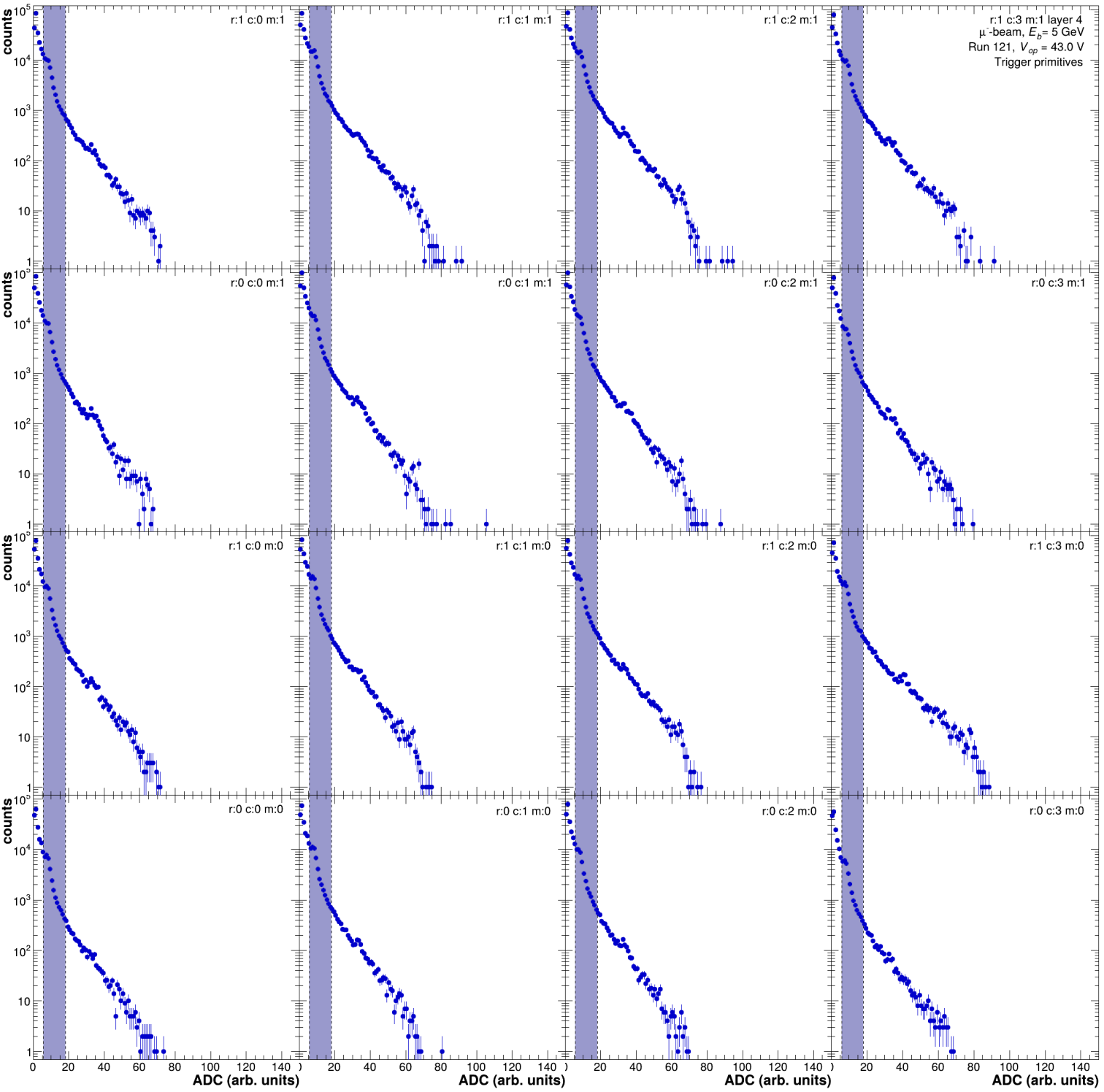

⟨ S/B signal ⟩ = 0.219
μ⁻-beam, E_b= 5 GeV, Run 121, V_{op} = 43.0 V
brd channel
layer
S/B signal region
10^{-1}
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14

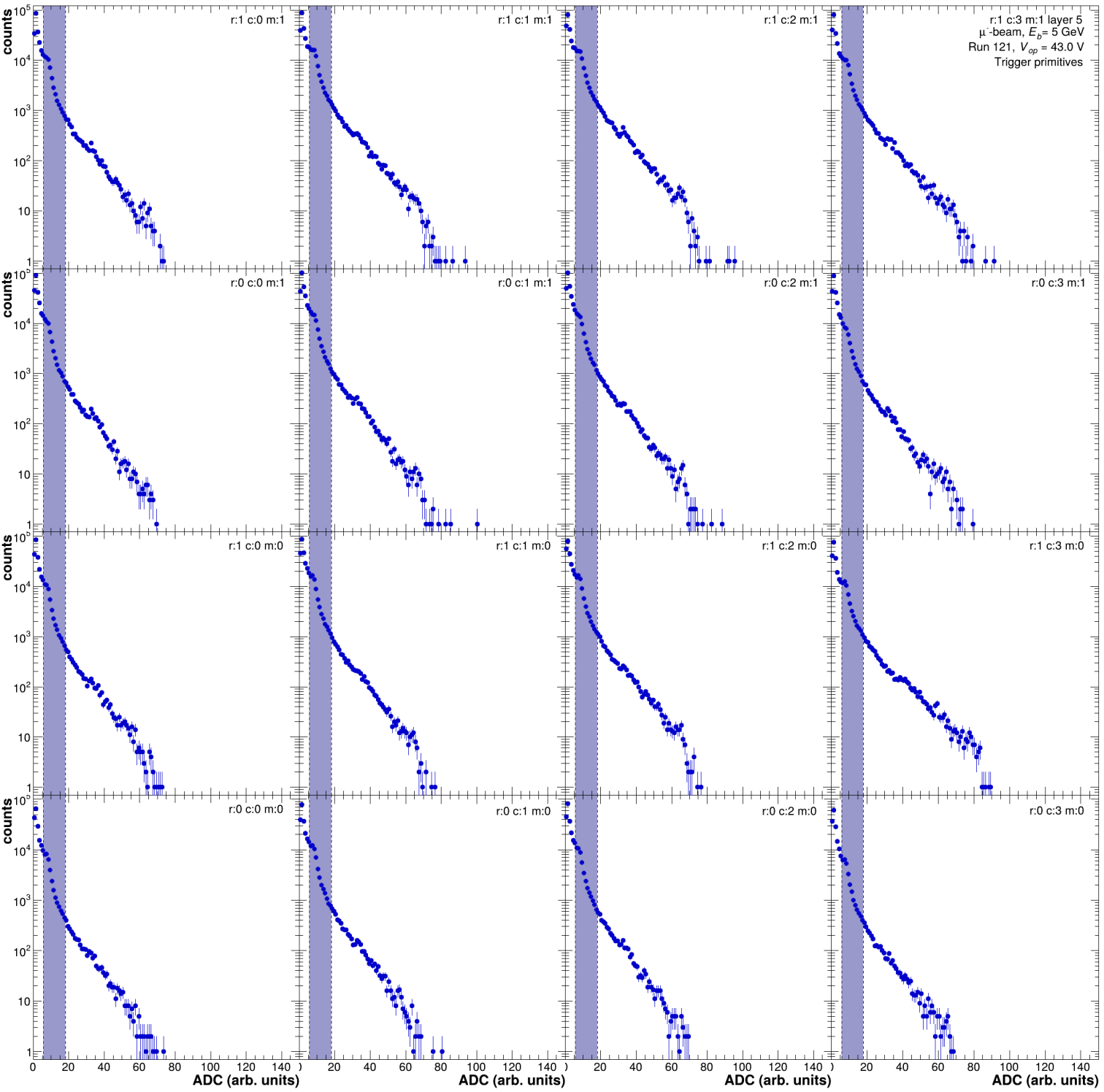

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 121, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

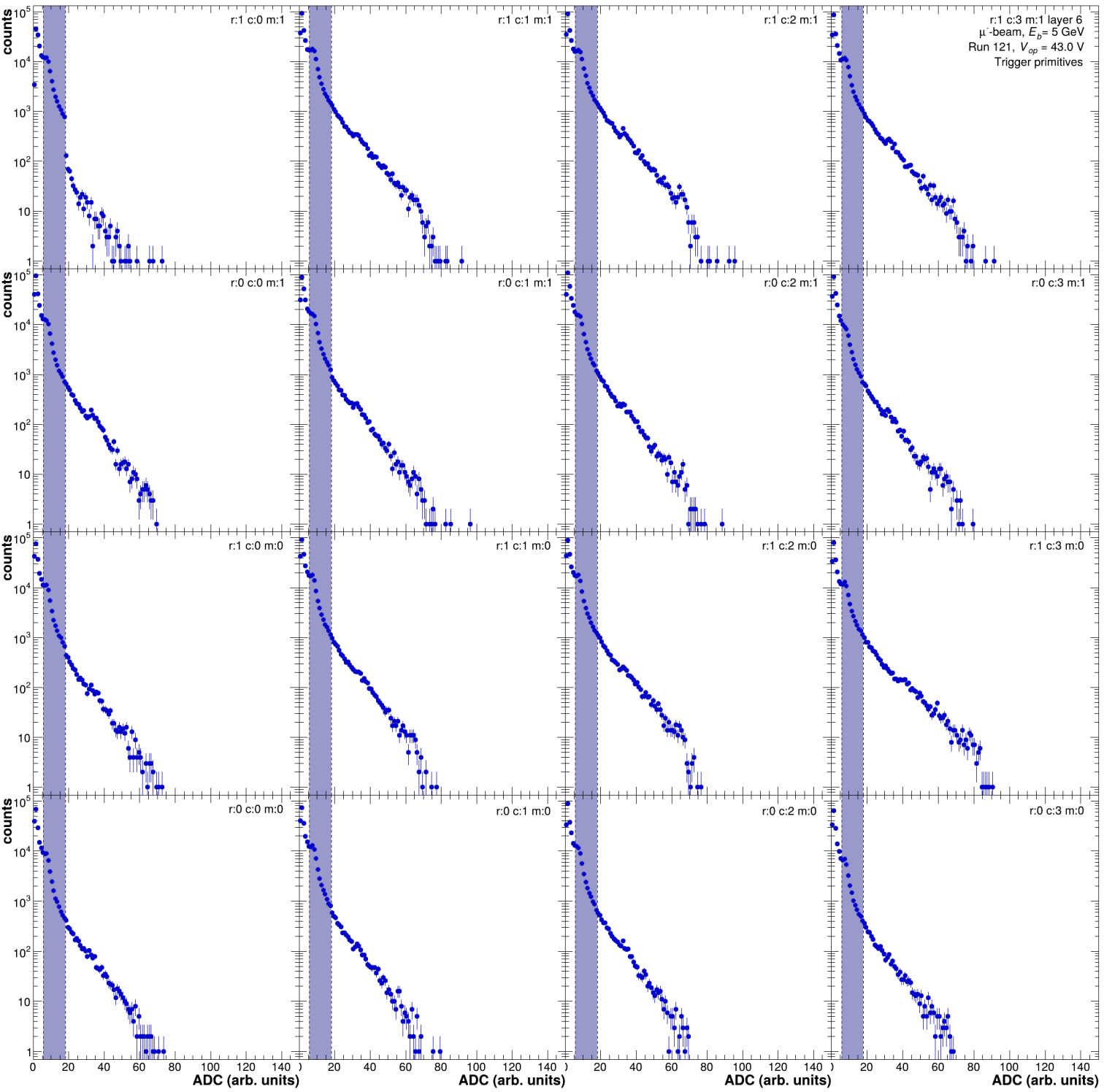

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 121, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

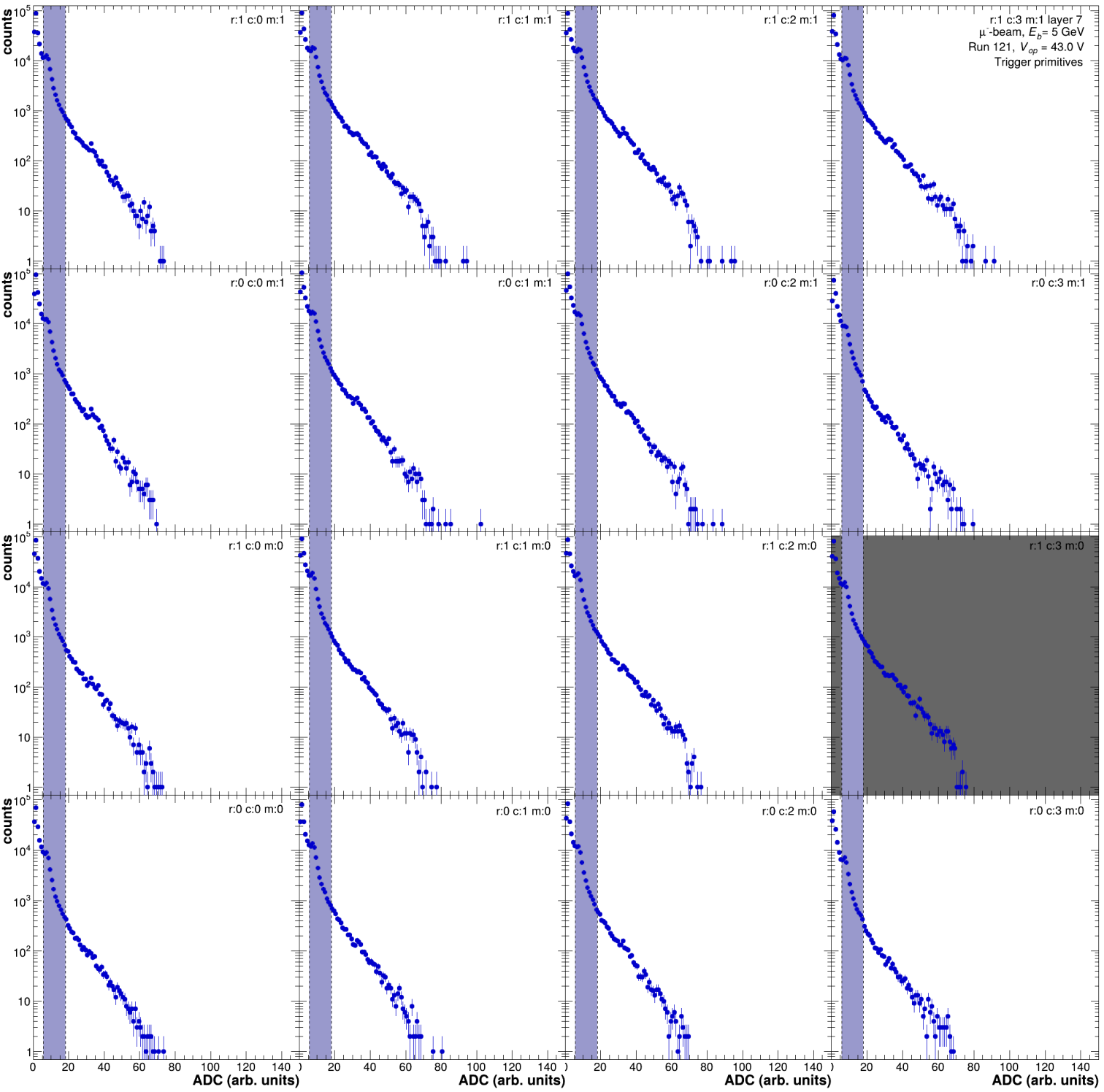

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 121, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 121, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 121, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 121, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 121, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 121, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)